

The Gradient of a Curve

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Differentiation (ref d1)

This document contains all 70 *Gradient of a Curve* questions, each with a fully worked solution and a TikZ diagram: a red **chord** with the limiting **tangent** (dashed orange) for the chord-gradient questions; the curve in blue with its **derivative** $f'(x)$ in teal for the first-principles results; and the curve with the **tangent** at the relevant point for the gradient/tangent questions.

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The Gradient of a Curve 01 (d1-001)

Difficulty: Foundation / Marks: 4

The curve C is given by $y = f(x)$ where $f(x) = x^2$.

Find the gradient of the chord joining the point where $x = 2$ to the point where:

(i) $x = 3$ (ii) $x = 2.5$ (iii) $x = 2.1$

What does this suggest about the gradient of the curve at $x = 2$?

Worked solution.

Step 1: Recall: gradient of chord from $x = 2$ to $x = a$ is $\frac{f(a) - f(2)}{a - 2}$.

$$\frac{a^2 - 4}{a - 2}$$

$$f(2) = 4.$$

Step 2: (i) $x = 3$.

$$\frac{9 - 4}{3 - 2} = \frac{5}{1} = 5$$

$$f(3) = 9.$$

Step 3: (ii) $x = 2.5$.

$$\frac{6.25 - 4}{2.5 - 2} = \frac{2.25}{0.5} = 4.5$$

$$f(2.5) = 6.25.$$

Step 4: (iii) $x = 2.1$.

$$\frac{4.41 - 4}{2.1 - 2} = \frac{0.41}{0.1} = 4.1$$

$$f(2.1) = 4.41.$$

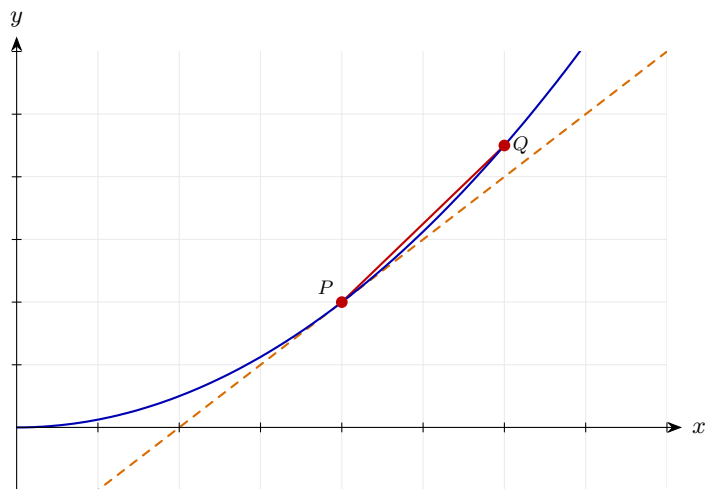
Step 5: As the second point gets closer to $x = 2$, the chord gradient approaches 4.

Gradient at $x = 2 \approx 4$

The gradient of the curve at $x = 2$ is 4. The diagram shows the three chords closing in on the tangent at $P(2, 4)$.

Final answer: (i) 5 (ii) 4.5 (iii) 4.1 The chord gradients approach 4, suggesting the gradient of the curve at $x = 2$ is 4.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 02 (d1-002)

Difficulty: Foundation / Marks: 4

The curve C is given by $y = x^2$. Find the gradient of the chord joining the point $(3, 9)$ to the point where:

(i) $x = 4$ (ii) $x = 3.5$ (iii) $x = 3.01$

Hence state the gradient of the curve at $x = 3$.

Worked solution.

Step 1: Chord gradient formula: $\frac{f(a) - f(3)}{a - 3} = \frac{a^2 - 9}{a - 3}$.

$$\frac{a^2 - 9}{a - 3}$$

$$f(3) = 9.$$

Step 2: (i) $x = 4$.

$$\frac{16 - 9}{4 - 3} = 7$$

$$f(4) = 16.$$

Step 3: (ii) $x = 3.5$.

$$\frac{12.25 - 9}{3.5 - 3} = \frac{3.25}{0.5} = 6.5$$

$$f(3.5) = 12.25.$$

Step 4: (iii) $x = 3.01$.

$$\frac{9.0601 - 9}{3.01 - 3} = \frac{0.0601}{0.01} = 6.01$$

$$f(3.01) = 9.0601.$$

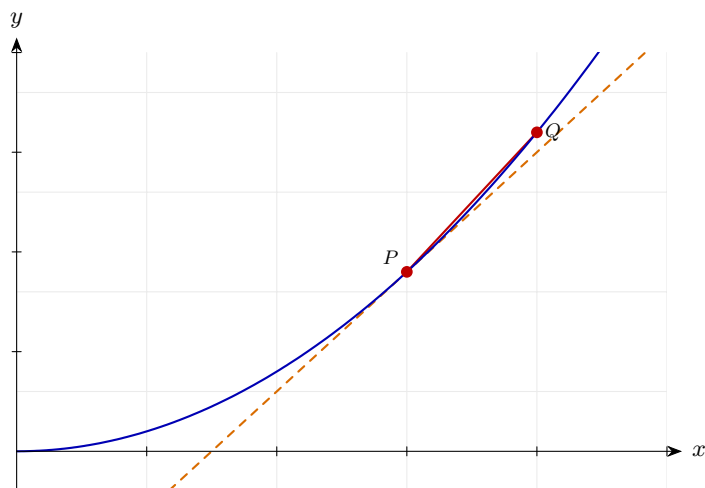
Step 5: The values approach 6.

Gradient at $x = 3 = 6$

As the second point approaches $x = 3$, the chord gradient approaches 6.

Final answer: (i) 7 (ii) 6.5 (iii) 6.01 Gradient at $x = 3$ is 6.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 03 (d1-003)

Difficulty: Foundation / Marks: 4

The curve C is given by $f(x) = x^3$. Find the gradient of the chord from $(2, 8)$ to the point where:

(i) $x = 3$ (ii) $x = 2.5$ (iii) $x = 2.1$

What value does the gradient appear to be approaching?

Worked solution.

Step 1: Chord gradient: $\frac{a^3 - 8}{a - 2}$.

$$\frac{a^3 - 8}{a - 2}$$

$$f(2) = 8.$$

Step 2: (i) $x = 3$.

$$\frac{27 - 8}{3 - 2} = 19$$

$$3^3 = 27.$$

Step 3: (ii) $x = 2.5$.

$$\frac{15.625 - 8}{2.5 - 2} = \frac{7.625}{0.5} = 15.25$$

$$2.5^3 = 15.625.$$

Step 4: (iii) $x = 2.1$.

$$\frac{9.261 - 8}{2.1 - 2} = \frac{1.261}{0.1} = 12.61$$

$$2.1^3 = 9.261.$$

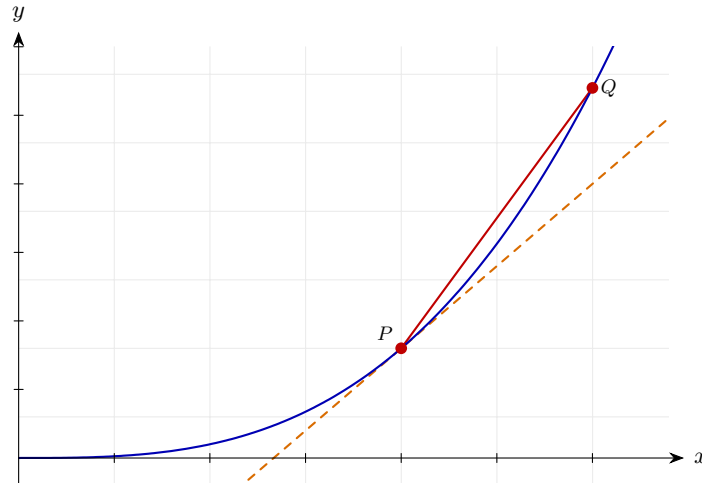
Step 5: The values approach 12.

Gradient at $x = 2 \approx 12$

This matches $3x^2|_{x=2} = 12$.

Final answer: (i)19 (ii)15.25 (iii)12.61 The gradient appears to approach 12.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 04 (d1-004)

Difficulty: Foundation / Marks: 4

The curve C is given by $f(x) = x^2 + 3x$. Find the gradient of the chord joining the point where $x = 1$ to the point where:

- (i) $x = 2$ (ii) $x = 1.5$ (iii) $x = 1.01$

Worked solution.

Step 1: Calculate $f(1)$.

$$f(1) = 1 + 3 = 4$$

Starting point is $(1, 4)$.

Step 2: (i) $f(2) = 4 + 6 = 10$.

$$\frac{10 - 4}{2 - 1} = 6$$

Chord gradient.

Step 3: (ii) $f(1.5) = 2.25 + 4.5 = 6.75$.

$$\frac{6.75 - 4}{1.5 - 1} = \frac{2.75}{0.5} = 5.5$$

Chord gradient.

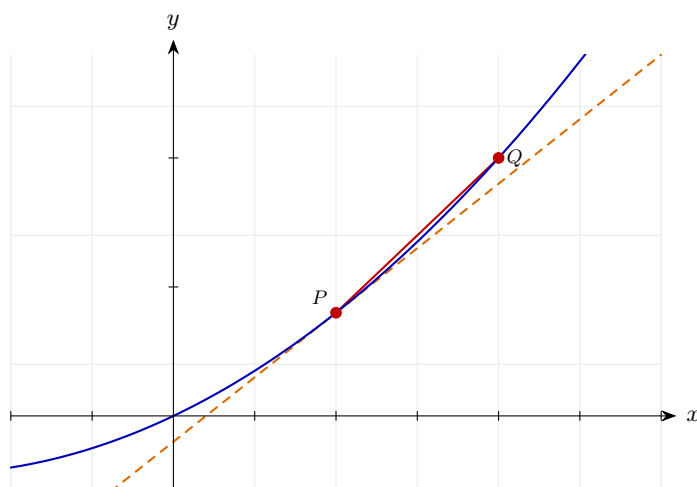
Step 4: (iii) $f(1.01) = 1.0201 + 3.03 = 4.0501$.

$$\frac{4.0501 - 4}{1.01 - 1} = \frac{0.0501}{0.01} = 5.01$$

Chord gradient approaching 5.

Final answer: (i) 6 (ii) 5.5 (iii) 5.01 The gradient at $x = 1$ appears to be 5.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 05 (d1-005)

Difficulty: Foundation / Marks: 3

The gradient of the curve $y = x^2$ at the point $(4, 16)$ is stated to be 8.

Verify this by finding the gradient of chords from $x = 4$ to $x = 5$, $x = 4.5$, and $x = 4.1$, and commenting on the pattern.

Worked solution.

Step 1: $x = 5$: $f(5) = 25$.

$$\frac{25 - 16}{5 - 4} = 9$$

Chord gradient.

Step 2: $x = 4.5$: $f(4.5) = 20.25$.

$$\frac{20.25 - 16}{4.5 - 4} = \frac{4.25}{0.5} = 8.5$$

Chord gradient.

Step 3: $x = 4.1$: $f(4.1) = 16.81$.

$$\frac{16.81 - 16}{4.1 - 4} = \frac{0.81}{0.1} = 8.1$$

Chord gradient.

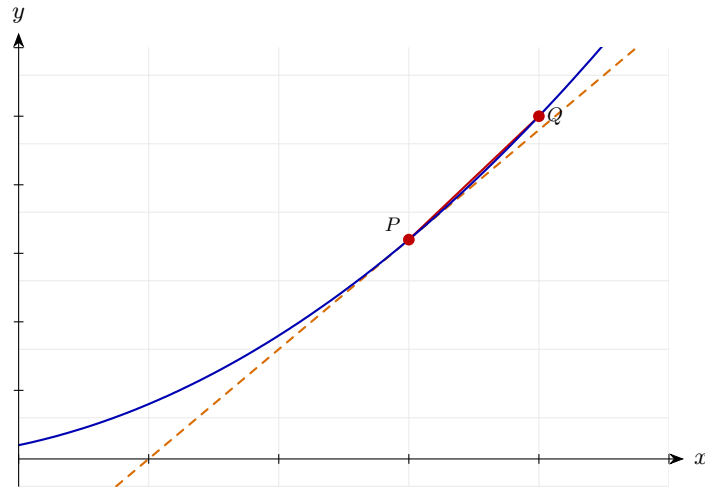
Step 4: Pattern: the chord gradients approach 8 as the second point approaches $x = 4$.

$$9 \rightarrow 8.5 \rightarrow 8.1 \rightarrow 8$$

This confirms the gradient at $x = 4$ is 8.

Final answer: Chord gradients are 9, 8.5, 8.1 — these approach 8, confirming the gradient at $x = 4$ is 8.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 06 (d1-006)

Difficulty: Foundation / Marks: 4

Differentiate $f(x) = x^2$ **from first principles to show that** $f'(x) = 2x$.

Worked solution.

Step 1: Write the first principles formula.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

This is the definition of the derivative.

Step 2: Substitute $f(x) = x^2$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$

Replace $f(x+h)$ with $(x+h)^2$.

Step 3: Expand $(x+h)^2$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$(x+h)^2 = x^2 + 2xh + h^2.$$

Step 4: Simplify the numerator.

$$f'(x) = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} (2x + h)$$

Cancel h from numerator and denominator (valid since $h \neq 0$).

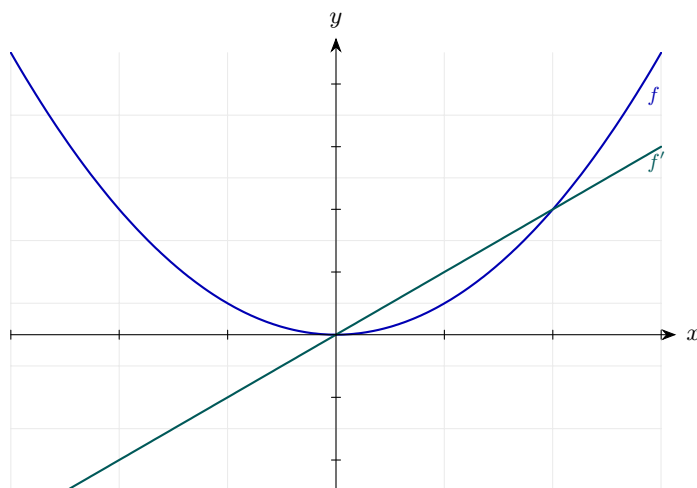
Step 5: Take the limit as $h \rightarrow 0$.

$$f'(x) = 2x$$

As $h \rightarrow 0$, $2x + h \rightarrow 2x$.

Final answer: $f'(x) = 2x$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 07 (d1-007)

Difficulty: Foundation / Marks: 4

Differentiate $f(x) = x^3$ from first principles.

Worked solution.

Step 1: Write the first principles formula.

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}$$

Definition of the derivative with $f(x) = x^3$.

Step 2: Expand $(x+h)^3$.

$$(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$$

Using the binomial expansion or repeated multiplication.

Step 3: Simplify the numerator.

$$\frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} = \frac{3x^2h + 3xh^2 + h^3}{h}$$

The x^3 terms cancel.

Step 4: Cancel h .

$$= 3x^2 + 3xh + h^2$$

Divide every term by h .

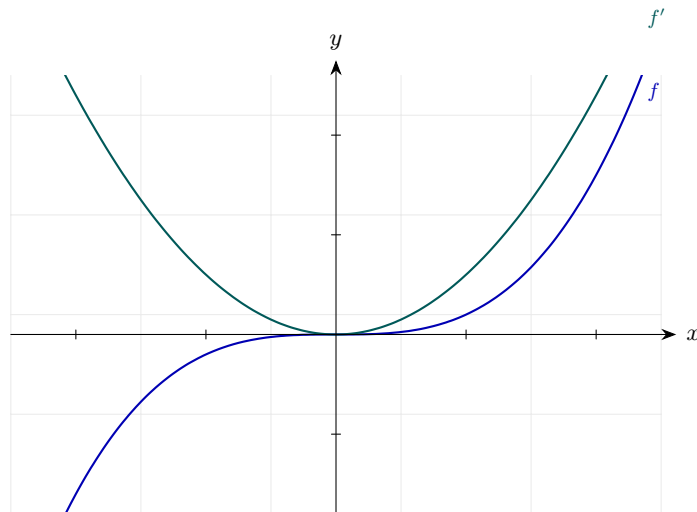
Step 5: Take the limit as $h \rightarrow 0$.

$$f'(x) = 3x^2$$

The terms involving h vanish.

Final answer: $f'(x) = 3x^2$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 08 (d1-008)

Difficulty: Foundation / Marks: 4

Differentiate $f(x) = 3x^2$ from first principles.

Worked solution.

Step 1: Apply the first principles formula.

$$f'(x) = \lim_{h \rightarrow 0} \frac{3(x+h)^2 - 3x^2}{h}$$

$$f(x+h) = 3(x+h)^2.$$

Step 2: Expand and simplify the numerator.

$$= \lim_{h \rightarrow 0} \frac{3(x^2 + 2xh + h^2) - 3x^2}{h} = \lim_{h \rightarrow 0} \frac{6xh + 3h^2}{h}$$

Expand the bracket then cancel $3x^2$.

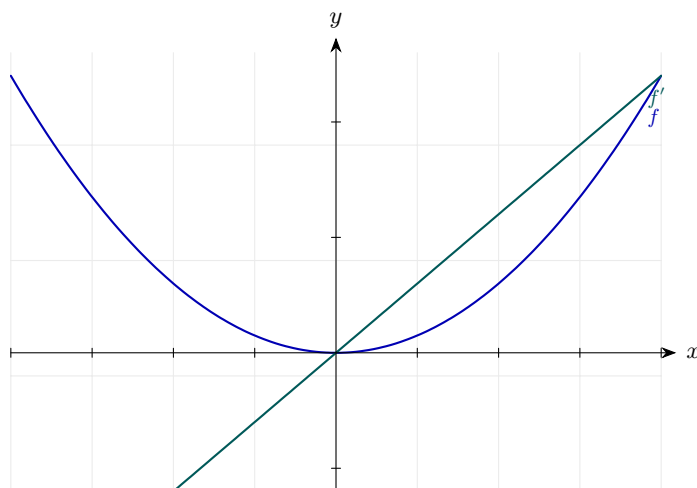
Step 3: Cancel h and take the limit.

$$= \lim_{h \rightarrow 0} (6x + 3h) = 6x$$

As $h \rightarrow 0$, the $3h$ term vanishes.

Final answer: $f'(x) = 6x$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 09 (d1-009)

Difficulty: Foundation / Marks: 4

Differentiate $f(x) = 2x^2 + 5$ from first principles.

Worked solution.

Step 1: Write $f(x + h)$.

$$f(x + h) = 2(x + h)^2 + 5 = 2x^2 + 4xh + 2h^2 + 5$$

Expand the bracket.

Step 2: Form the difference quotient.

$$\frac{f(x + h) - f(x)}{h} = \frac{(2x^2 + 4xh + 2h^2 + 5) - (2x^2 + 5)}{h} = \frac{4xh + 2h^2}{h}$$

The $2x^2$ and 5 terms cancel.

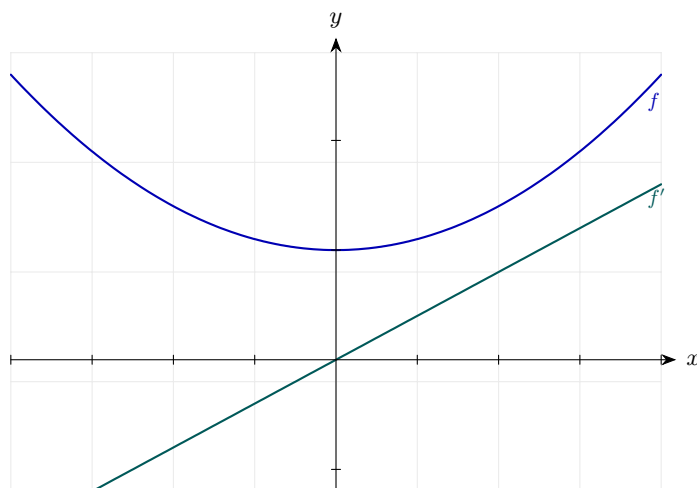
Step 3: Simplify and take the limit.

$$= 4x + 2h \xrightarrow{h \rightarrow 0} 4x$$

Cancel h , then let $h \rightarrow 0$.

Final answer: $f'(x) = 4x$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 10 (d1-010)

Difficulty: Foundation / Marks: 3

Differentiate $y = 5x + 3$ from first principles.

Worked solution.

Step 1: Write $f(x + h)$.

$$f(x + h) = 5(x + h) + 3 = 5x + 5h + 3$$

Substitute $x + h$ for x .

Step 2: Form the difference quotient.

$$\frac{f(x + h) - f(x)}{h} = \frac{(5x + 5h + 3) - (5x + 3)}{h} = \frac{5h}{h} = 5$$

The $5x$ and 3 terms cancel, leaving just 5 .

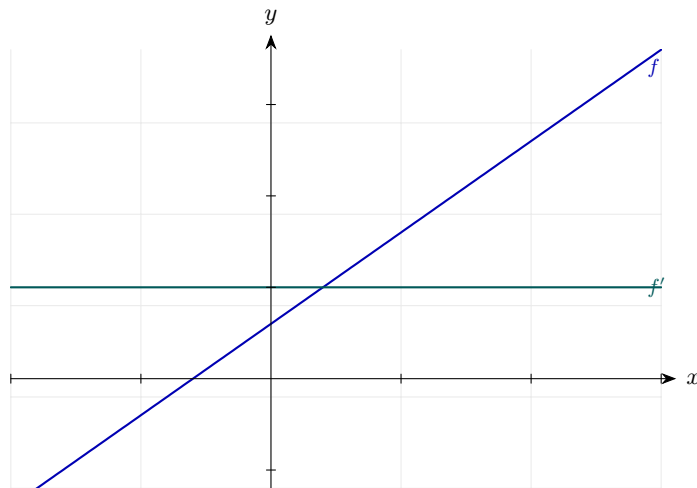
Step 3: Take the limit.

$$\lim_{h \rightarrow 0} 5 = 5$$

The result is already constant — the gradient of a straight line is constant, as expected.

Final answer: $\frac{dy}{dx} = 5$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 11 (d1-011)

Difficulty: Foundation / Marks: 4

Differentiate $f(x) = x^2 - x$ from first principles.

Worked solution.

Step 1: Write $f(x + h)$.

$$f(x + h) = (x + h)^2 - (x + h) = x^2 + 2xh + h^2 - x - h$$

Expand each term.

Step 2: Form the difference quotient.

$$\frac{f(x + h) - f(x)}{h} = \frac{(x^2 + 2xh + h^2 - x - h) - (x^2 - x)}{h} = \frac{2xh + h^2 - h}{h}$$

The x^2 and $-x$ terms cancel.

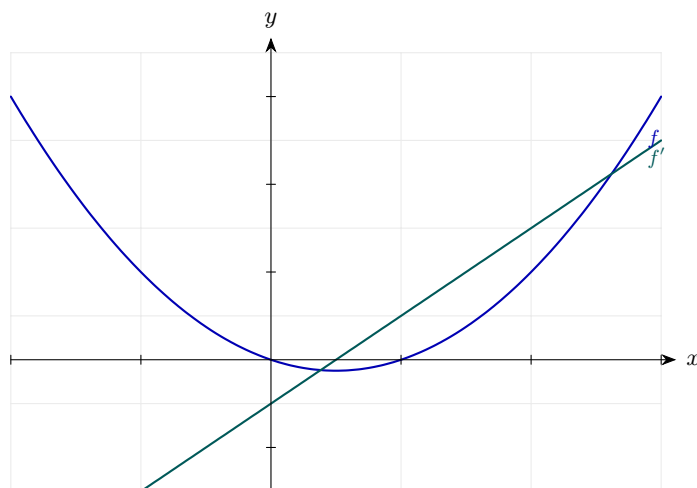
Step 3: Cancel h and take the limit.

$$= 2x + h - 1 \xrightarrow{h \rightarrow 0} 2x - 1$$

The h term vanishes as $h \rightarrow 0$.

Final answer: $f'(x) = 2x - 1$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 12 (d1-012)

Difficulty: Foundation / Marks: 4

Differentiate $f(x) = 2x^2 + 3x$ from first principles.

Worked solution.

Step 1: Write $f(x + h)$.

$$f(x + h) = 2(x + h)^2 + 3(x + h) = 2x^2 + 4xh + 2h^2 + 3x + 3h$$

Expand $2(x + h)^2$ and $3(x + h)$ separately.

Step 2: Subtract $f(x) = 2x^2 + 3x$.

$$f(x + h) - f(x) = 4xh + 2h^2 + 3h$$

The $2x^2$ and $3x$ terms cancel.

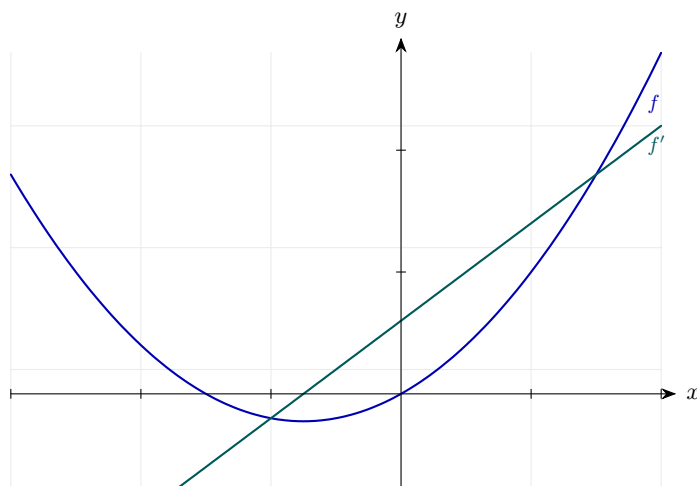
Step 3: Divide by h and take the limit.

$$\frac{4xh + 2h^2 + 3h}{h} = 4x + 2h + 3 \xrightarrow{h \rightarrow 0} 4x + 3$$

Cancel h , then let $h \rightarrow 0$.

Final answer: $f'(x) = 4x + 3$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 13 (d1-013)

Difficulty: Foundation / Marks: 5

Differentiate $y = 3x^2 - 4x + 1$ from first principles.

Worked solution.

Step 1: Write $f(x + h)$.

$$f(x + h) = 3(x + h)^2 - 4(x + h) + 1 = 3x^2 + 6xh + 3h^2 - 4x - 4h + 1$$

Expand each term carefully.

Step 2: Subtract $f(x) = 3x^2 - 4x + 1$.

$$f(x + h) - f(x) = 6xh + 3h^2 - 4h$$

The $3x^2$, $-4x$ and $+1$ terms all cancel.

Step 3: Divide by h .

$$\frac{6xh + 3h^2 - 4h}{h} = 6x + 3h - 4$$

Divide every term by h .

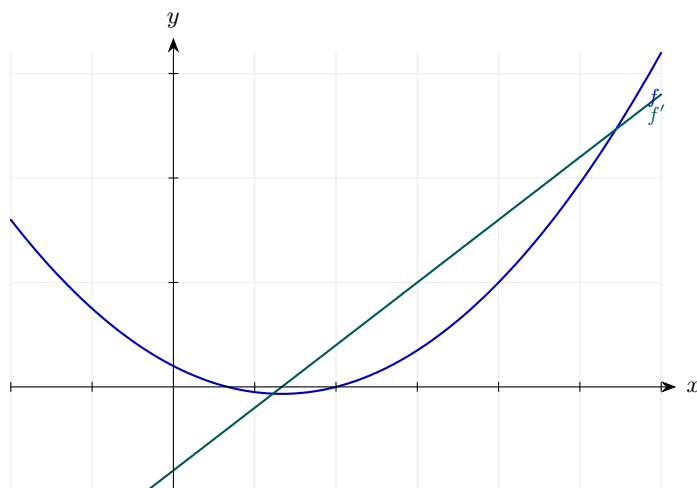
Step 4: Take the limit.

$$\lim_{h \rightarrow 0} (6x + 3h - 4) = 6x - 4$$

The $3h$ term vanishes.

Final answer: $\frac{dy}{dx} = 6x - 4$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 14 (d1-014)

Difficulty: Foundation / Marks: 4

Differentiate $f(x) = x^2 + 4x - 7$ from first principles.

Worked solution.

Step 1: Write $f(x + h)$.

$$f(x + h) = (x + h)^2 + 4(x + h) - 7 = x^2 + 2xh + h^2 + 4x + 4h - 7$$

Expand all brackets.

Step 2: Find $f(x + h) - f(x)$.

$$= 2xh + h^2 + 4h$$

The x^2 , $4x$ and -7 terms cancel.

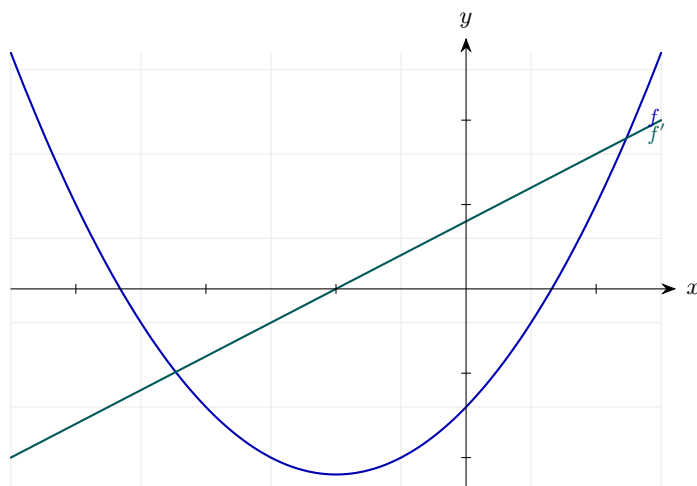
Step 3: Divide by h and take the limit.

$$= 2x + h + 4 \xrightarrow{h \rightarrow 0} 2x + 4$$

The h term vanishes.

Final answer: $f'(x) = 2x + 4$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 15 (d1-015)

Difficulty: Foundation / Marks: 4

By differentiating $f(x) = x^2$ from first principles, find the gradient of the curve at the point $(5, 25)$.

Worked solution.

Step 1: Differentiate $f(x) = x^2$ from first principles.

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = 2x$$

Standard first-principles derivation.

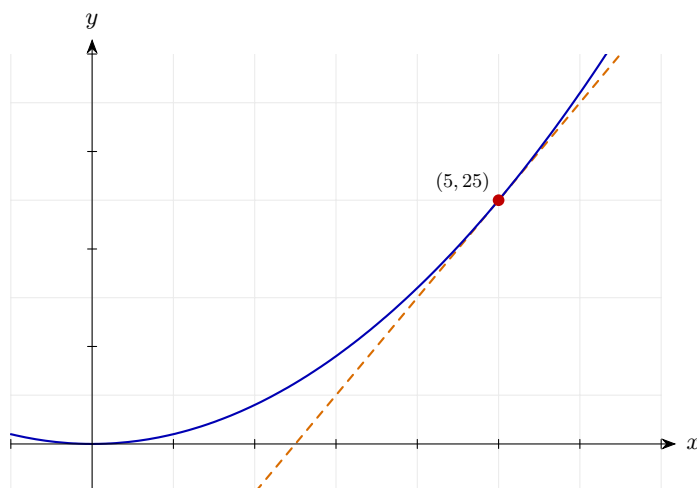
Step 2: Substitute $x = 5$.

$$f'(5) = 2 \times 5 = 10$$

Replace x with 5 in the derivative. The diagram shows the tangent with gradient 10 touching the curve at $(5, 25)$.

Final answer: Gradient at $(5, 25)$ is 10.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 16 (d1-016)

Difficulty: Foundation / Marks: 5

By differentiating $f(x) = x^3$ from first principles, find the gradient of the curve at $x = -2$.

Worked solution.

Step 1: Differentiate from first principles.

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}$$

Definition of derivative.

Step 2: Expand $(x+h)^3$ and simplify.

$$= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h} = \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) = 3x^2$$

Cancel h and take the limit.

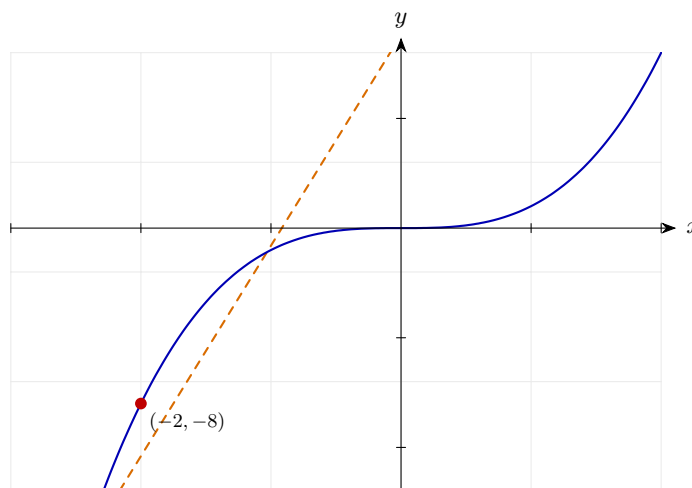
Step 3: Substitute $x = -2$.

$$f'(-2) = 3(-2)^2 = 3 \times 4 = 12$$

The gradient is positive here.

Final answer: Gradient at $x = -2$ is 12.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 17 (d1-017)

Difficulty: Foundation / Marks: 4

By differentiating $f(x) = 2x^2$ from first principles, find the gradient of the tangent to the curve at the point $(-3, 18)$.

Worked solution.

Step 1: Differentiate from first principles.

$$f'(x) = \lim_{h \rightarrow 0} \frac{2(x+h)^2 - 2x^2}{h} = \lim_{h \rightarrow 0} \frac{4xh + 2h^2}{h} = 4x$$

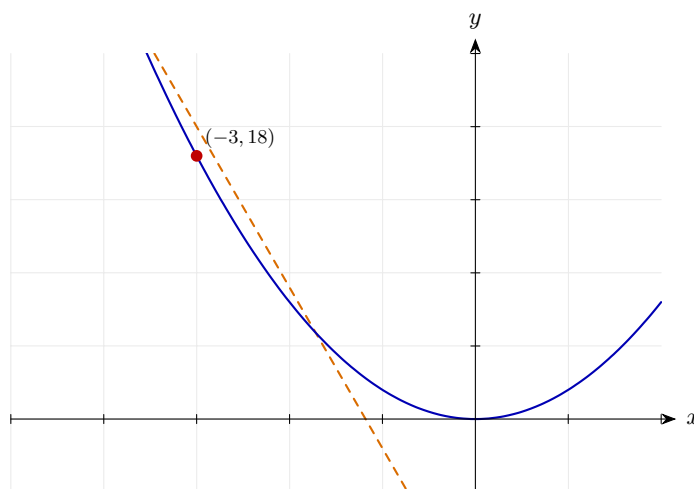
Cancel h and take the limit.

Step 2: Evaluate at $x = -3$.

$$f'(-3) = 4(-3) = -12$$

The negative gradient means the curve is decreasing at this point.

Final answer: Gradient of the tangent at $(-3, 18)$ is -12 .

Diagram.

Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 18 (d1-018)

Difficulty: Foundation / Marks: 4

By differentiating from first principles, find the gradient of $y = x^2 - 3x$ at the point where $x = 4$.

Worked solution.

Step 1: Find $f(x+h) - f(x)$.

$$[(x+h)^2 - 3(x+h)] - [x^2 - 3x] = 2xh + h^2 - 3h$$

Expand and cancel.

Step 2: Divide by h and take the limit.

$$2x + h - 3 \xrightarrow{h \rightarrow 0} 2x - 3$$

So $f'(x) = 2x - 3$.

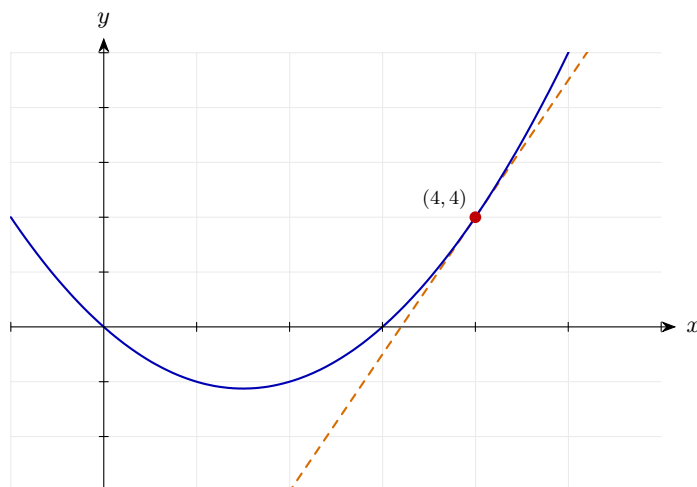
Step 3: Substitute $x = 4$.

$$f'(4) = 2(4) - 3 = 5$$

Evaluate the derivative at the given point.

Final answer: Gradient at $x = 4$ is 5.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 19 (d1-019)

Difficulty: Foundation / Marks: 4

The curve C has equation $y = 4x^2 - 1$. By differentiating from first principles, find the gradient at the point $(2, 15)$.

Worked solution.

Step 1: Find $f(x + h) - f(x)$.

$$[4(x + h)^2 - 1] - [4x^2 - 1] = 4(x^2 + 2xh + h^2) - 4x^2 = 8xh + 4h^2$$

The $4x^2$ and -1 terms cancel.

Step 2: Divide by h and take the limit.

$$8x + 4h \xrightarrow{h \rightarrow 0} 8x$$

So $f'(x) = 8x$.

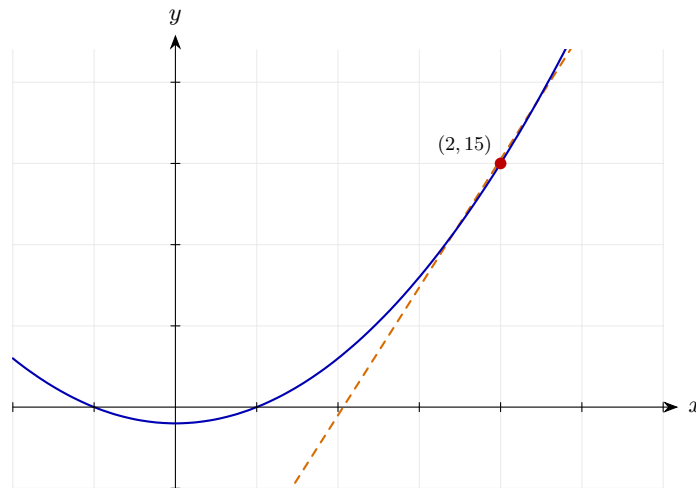
Step 3: Substitute $x = 2$.

$$f'(2) = 8 \times 2 = 16$$

Gradient at $(2, 15)$ is 16.

Final answer: Gradient at $(2, 15)$ is 16.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 20 (d1-020)

Difficulty: Foundation / Marks: 5

The curve C has equation $y = x^2 + 6$.

(a) Differentiate from first principles to find $\frac{dy}{dx}$.

(b) Find the coordinates of the point on C where the gradient is 8.

Worked solution.

Step 1: (a) Apply first principles to $y = x^2 + 6$.

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 + 6 - x^2 - 6}{h} = \frac{2xh + h^2}{h} = 2x + h \xrightarrow{h \rightarrow 0} 2x$$

The constant 6 cancels out.

Step 2: (b) Set $\frac{dy}{dx} = 8$.

$$2x = 8 \Rightarrow x = 4$$

Solve for x .

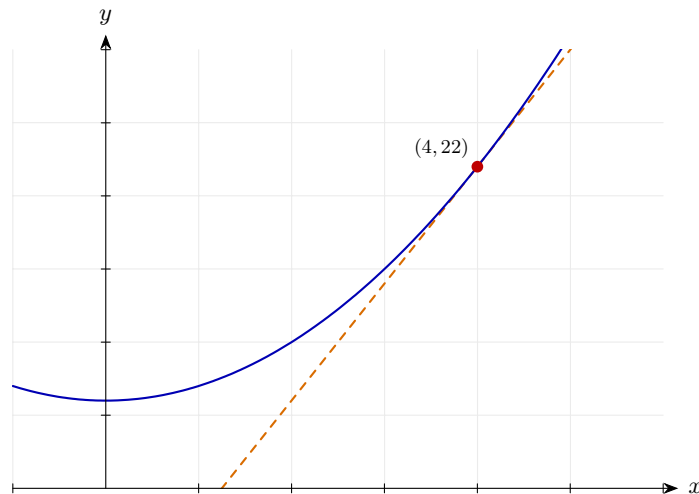
Step 3: Find the y -coordinate.

$$y = 4^2 + 6 = 22$$

Substitute $x = 4$ into the original equation.

Final answer: (a) $\frac{dy}{dx} = 2x$ (b) (4, 22)

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 21 (d1-021)

Difficulty: Foundation / Marks: 5

The curve C has equation $y = 3x^2$.

(a) Find $\frac{dy}{dx}$ by differentiating from first principles.

(b) Find the coordinates of the point on C where the gradient is -18 .

Worked solution.

Step 1: (a) From first principles.

$$\frac{3(x+h)^2 - 3x^2}{h} = \frac{6xh + 3h^2}{h} = 6x + 3h \xrightarrow{h \rightarrow 0} 6x$$

Standard calculation.

Step 2: (b) Set $6x = -18$.

$$x = -3$$

Divide both sides by 6.

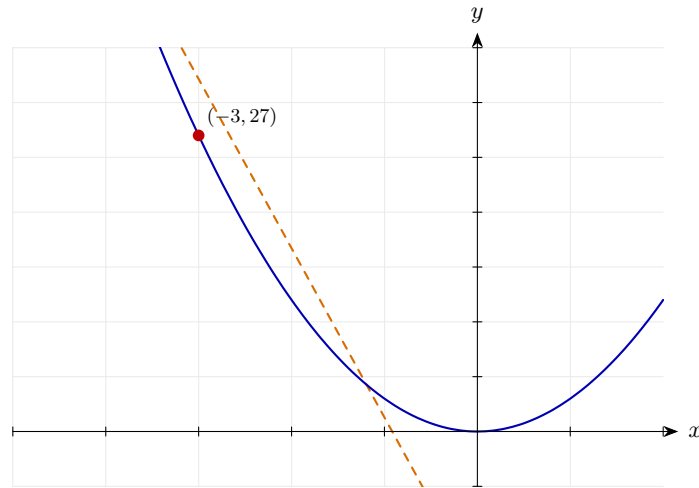
Step 3: Find y .

$$y = 3(-3)^2 = 27$$

Substitute $x = -3$.

Final answer: (a) $\frac{dy}{dx} = 6x$ (b) $(-3, 27)$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 22 (d1-022)

Difficulty: Foundation / Marks: 5

The curve C has equation $y = x^2 - 5x$.

- (a) Differentiate from first principles to find $\frac{dy}{dx}$.
 (b) Find the x -value where the tangent to C is parallel to the x -axis.

Worked solution.

Step 1: (a) From first principles.

$$f(x+h) - f(x) = [(x+h)^2 - 5(x+h)] - [x^2 - 5x] = 2xh + h^2 - 5h$$

Expand and simplify.

Step 2: Divide by h and take the limit.

$$2x + h - 5 \xrightarrow{h \rightarrow 0} 2x - 5$$

Derivative is $2x - 5$.

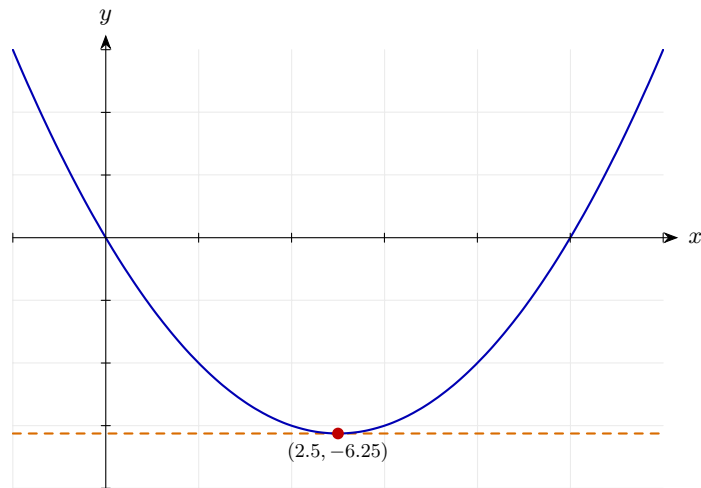
Step 3: (b) A tangent parallel to the x -axis has gradient 0.

$$2x - 5 = 0 \Rightarrow x = 2.5$$

Solve $\frac{dy}{dx} = 0$.

Final answer: (a) $\frac{dy}{dx} = 2x - 5$ (b) $x = 2.5$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 23 (d1-023)

Difficulty: Foundation / Marks: 5

The curve C has equation $y = x^3$.

- (a) Differentiate from first principles to show $\frac{dy}{dx} = 3x^2$.
 (b) Find the two values of x where the gradient of C is 27.

Worked solution.

Step 1: (a) From first principles (see d1-007 for full working).

$$\frac{dy}{dx} = 3x^2$$

Standard first-principles result for $y = x^3$.

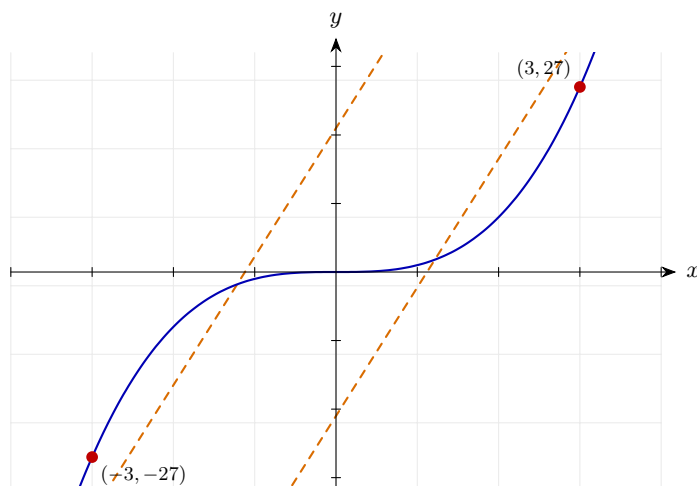
Step 2: (b) Set $3x^2 = 27$.

$$x^2 = 9 \Rightarrow x = \pm 3$$

Divide by 3 then square-root, giving two solutions.

Final answer: (b) $x = 3$ or $x = -3$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 24 (d1-024)

Difficulty: Foundation / Marks: 5

The curve C has equation $y = x^2 + 3$. A tangent to C has gradient 10.

(a) Differentiate $y = x^2 + 3$ from first principles.

(b) Find the equation of the tangent.

Worked solution.

Step 1: (a) From first principles.

$$\frac{(x+h)^2 + 3 - x^2 - 3}{h} = \frac{2xh + h^2}{h} \xrightarrow{h \rightarrow 0} 2x$$

Constant 3 cancels out.

Step 2: (b) Set $2x = 10$ to find the point of tangency.

$$x = 5 \Rightarrow y = 25 + 3 = 28$$

Point of tangency is $(5, 28)$.

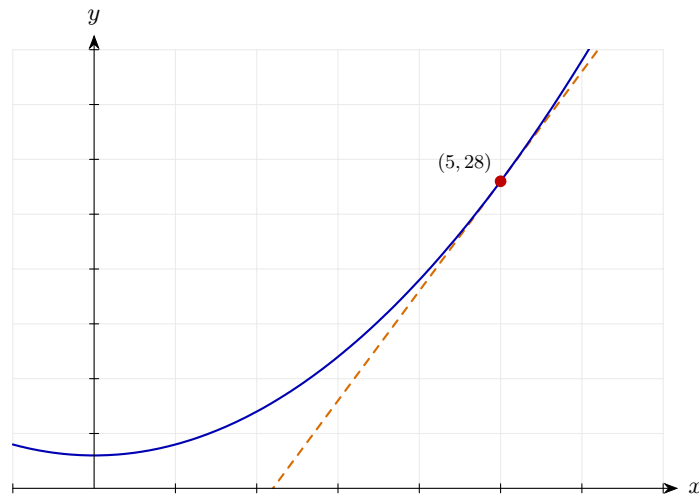
Step 3: Use $y - y_1 = m(x - x_1)$.

$$y - 28 = 10(x - 5) \Rightarrow y = 10x - 22$$

Gradient $m = 10$, point $(5, 28)$.

Final answer: (a) $\frac{dy}{dx} = 2x$ (b) $y = 10x - 22$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 25 (d1-025)

Difficulty: Foundation / Marks: 3

Explain what $f'(x)$ represents geometrically for the curve $y = f(x)$, and describe what it means when $f'(x) > 0$, $f'(x) = 0$, and $f'(x) < 0$.

Worked solution.

Step 1: $f'(x)$ is the gradient of the curve at the point $(x, f(x))$.

$$f'(x) = \text{gradient of tangent at } x$$

It measures the instantaneous rate of change of y with respect to x .

Step 2: $f'(x) > 0$: the curve is increasing (sloping upward) at that point.

$$f'(x) > 0 \Rightarrow \text{curve is increasing}$$

Moving left to right, y is getting larger.

Step 3: $f'(x) = 0$: the tangent is horizontal — the curve has a stationary point.

$$f'(x) = 0 \Rightarrow \text{stationary point (turning point or point of inflection)}$$

The curve is momentarily neither increasing nor decreasing.

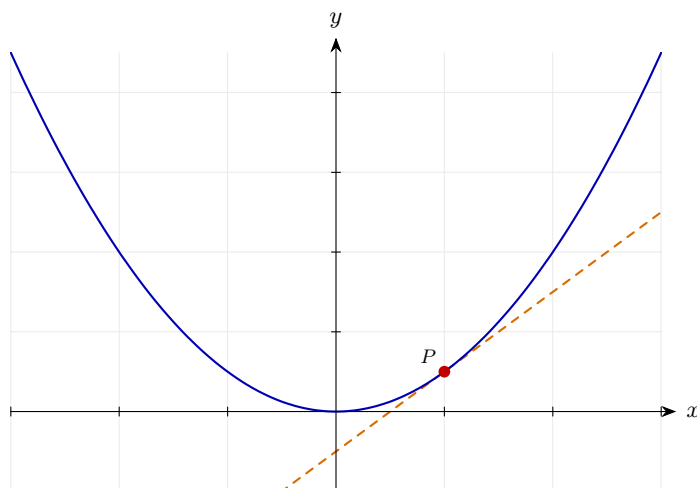
Step 4: $f'(x) < 0$: the curve is decreasing at that point.

$$f'(x) < 0 \Rightarrow \text{curve is decreasing}$$

Moving left to right, y is getting smaller.

Final answer: $f'(x)$ is the gradient of the tangent. Positive: increasing; zero: stationary; negative: decreasing.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 26 (d1-026)

Difficulty: Foundation / Marks: 4

The function $f(x) = x^2 - 4x + 1$ has derivative $f'(x) = 2x - 4$.

(a) Find the value of x where the gradient is zero.

(b) Determine whether the function is increasing or decreasing at $x = 0$ and $x = 4$.

Worked solution.

Step 1: (a) Set $f'(x) = 0$.

$$2x - 4 = 0 \Rightarrow x = 2$$

The curve has a stationary point at $x = 2$.

Step 2: (b) At $x = 0$.

$$f'(0) = 2(0) - 4 = -4 < 0$$

The function is decreasing at $x = 0$.

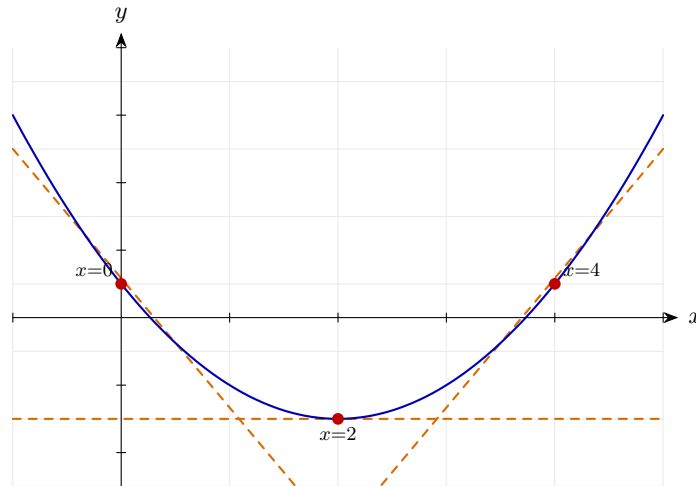
Step 3: At $x = 4$.

$$f'(4) = 2(4) - 4 = 4 > 0$$

The function is increasing at $x = 4$.

Final answer: (a) $x = 2$ (b) Decreasing at $x = 0$; increasing at $x = 4$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 27 (d1-027)

Difficulty: Foundation / Marks: 3

Explain the difference between the gradient of a chord and the gradient of a tangent to a curve at a point, and how differentiation from first principles connects the two.

Worked solution.

Step 1: A chord is a straight line joining two distinct points on a curve.

$$\text{Gradient of chord} = \frac{f(x+h) - f(x)}{h}$$

This is an average rate of change over an interval of width h .

Step 2: A tangent touches the curve at exactly one point, giving the instantaneous rate of change.

$$\text{Gradient of tangent} = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

As the two points on the chord are brought infinitely close together ($h \rightarrow 0$), the chord becomes the tangent.

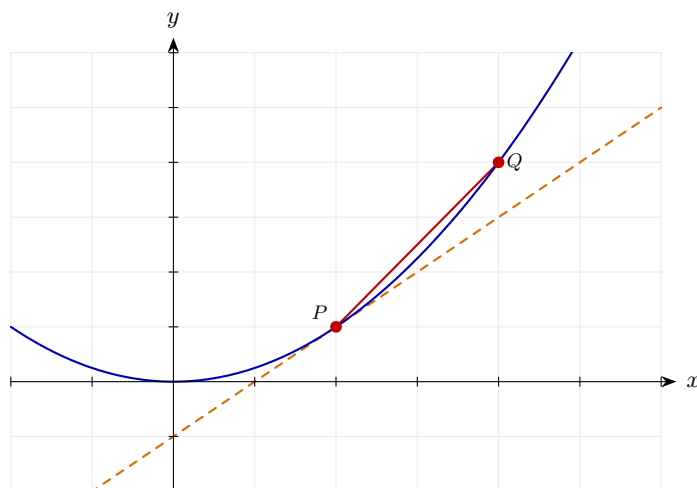
Step 3: First principles shows that the tangent gradient is the limit of the chord gradients.

$$\text{Chord gradient} \xrightarrow{h \rightarrow 0} \text{Tangent gradient}$$

Differentiating from first principles formalises this limiting process.

Final answer: A chord connects two points (average rate of change); a tangent touches at one point (instantaneous rate). First principles takes the limit as the chord length shrinks to zero, turning the chord gradient into the tangent gradient.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 28 (d1-028)

Difficulty: Foundation / Marks: 4

The area $A \text{ cm}^2$ of a square with side $x \text{ cm}$ is given by $A = x^2$.

By differentiating from first principles, find $\frac{dA}{dx}$ and interpret its meaning.

Worked solution.

Step 1: Apply first principles to $A = x^2$.

$$\frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \xrightarrow{h \rightarrow 0} 2x$$

Standard first-principles derivation.

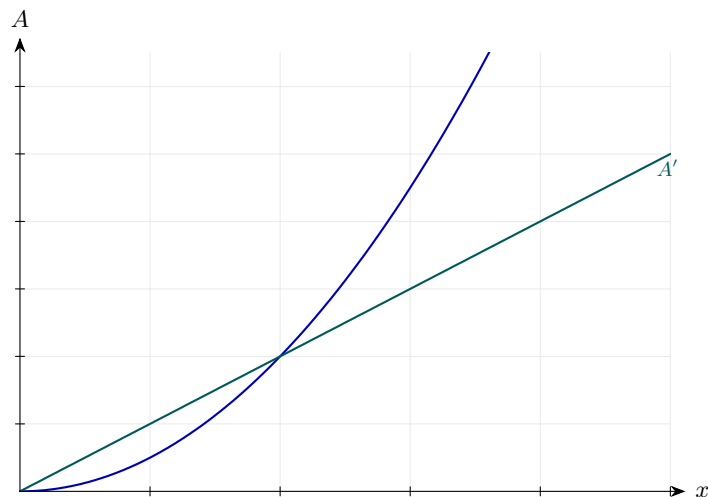
Step 2: Interpret the result.

$$\frac{dA}{dx} = 2x$$

$\frac{dA}{dx} = 2x$ represents the rate at which the area increases as the side length increases. At any side length x , the area increases by approximately $2x \text{ cm}^2$ for each additional cm of side length.

Final answer: $\frac{dA}{dx} = 2x$. It represents the rate of increase of area per unit increase in side length — numerically equal to the perimeter of two sides.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 29 (d1-029)

Difficulty: Foundation / Marks: 3

Show, by differentiating from first principles, that the derivative of any constant $f(x) = c$ is zero.

Worked solution.

Step 1: Write the first principles formula for $f(x) = c$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h}$$

Since $f(x) = c$ for all x , $f(x+h) = c$ as well.

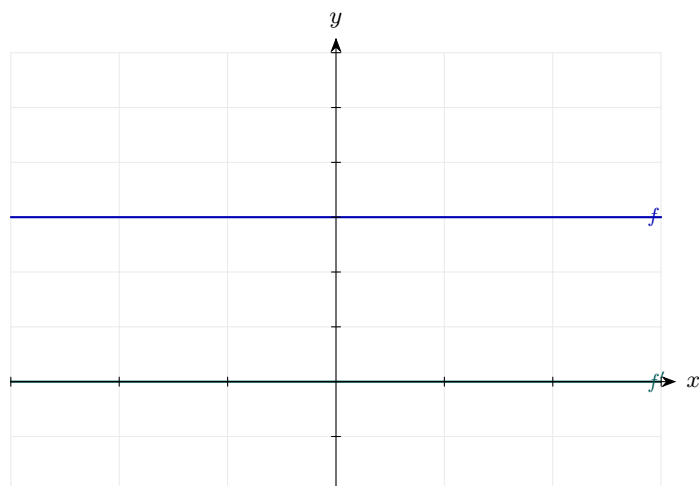
Step 2: Simplify.

$$= \lim_{h \rightarrow 0} \frac{0}{h} = \lim_{h \rightarrow 0} 0 = 0$$

Zero divided by anything non-zero is zero.

Final answer: $f'(x) = 0$ for any constant $f(x) = c$. This makes sense — a horizontal line has zero gradient.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 30 (d1-030)

Difficulty: Foundation / Marks: 3

Differentiate $f(x) = x$ from first principles and comment on the result.

Worked solution.

Step 1: Write the first principles formula.

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = \lim_{h \rightarrow 0} 1 = 1$$

The x terms cancel, leaving $h/h = 1$.

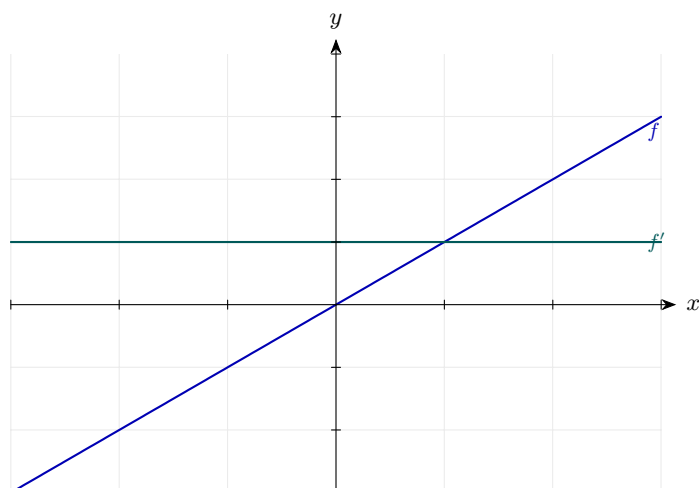
Step 2: Comment.

$$f'(x) = 1 \text{ for all } x$$

This is expected: $y = x$ is a straight line with gradient 1 everywhere.

Final answer: $f'(x) = 1$. The gradient of $y = x$ is constant at 1 for all x , which matches the slope of the line.

Diagram.



The Gradient of a Curve 31 (d1-031)

Difficulty: Foundation / Marks: 7

- (a) Differentiate $f(x) = x^2 + 2x - 3$ from first principles.
(b) Hence find the gradient of the curve at the point where $x = -1$.
(c) Find the equation of the tangent to the curve at this point.

Worked solution.

Step 1: (a) Find $f(x+h) - f(x)$.

$$[(x+h)^2 + 2(x+h) - 3] - [x^2 + 2x - 3] = 2xh + h^2 + 2h$$

Expand and cancel terms.

Step 2: Divide by h and take the limit.

$$2x + h + 2 \xrightarrow{h \rightarrow 0} 2x + 2$$

So $f'(x) = 2x + 2$.

Step 3: (b) Substitute $x = -1$.

$$f'(-1) = 2(-1) + 2 = 0$$

The gradient at $x = -1$ is 0 — this is a stationary point.

Step 4: (c) Find the y -coordinate.

$$f(-1) = 1 - 2 - 3 = -4$$

Point of tangency is $(-1, -4)$.

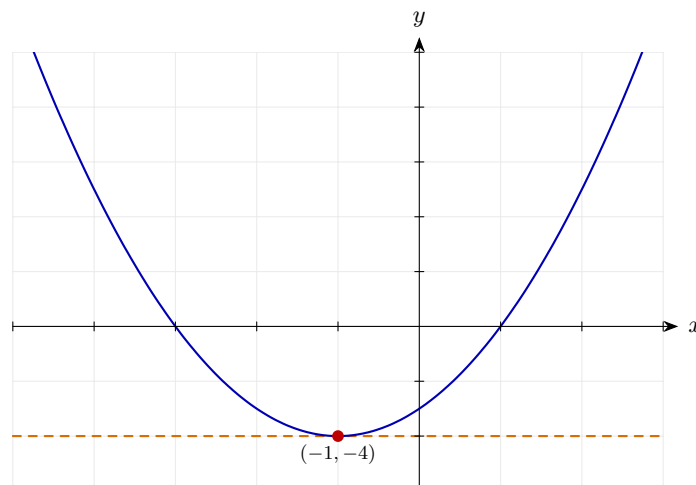
Step 5: Write the tangent equation (horizontal line since gradient = 0).

$$y = -4$$

A tangent with gradient 0 at $(-1, -4)$ is the horizontal line $y = -4$.

Final answer: (a) $f'(x) = 2x + 2$ (b) Gradient = 0 (c) $y = -4$

Diagram.



The Gradient of a Curve 32 (d1-032)

Difficulty: Foundation / Marks: 5

(a) Show, by differentiating $f(x) = x^2$ from first principles, that $f'(x) = 2x$.

(b) The point P lies on the curve $y = x^2$. The gradient of the curve at P is -6 . Find the coordinates of P .

Worked solution.

Step 1: (a) From first principles.

$$\frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h \xrightarrow{h \rightarrow 0} 2x \checkmark$$

This confirms $f'(x) = 2x$.

Step 2: (b) Set $2x = -6$.

$$x = -3$$

Solve for x .

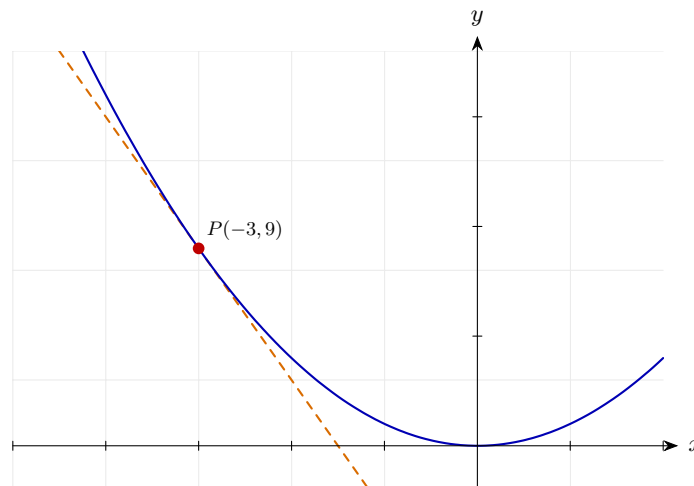
Step 3: Find y .

$$y = (-3)^2 = 9$$

Coordinates of P are $(-3, 9)$.

Final answer: (b) $P = (-3, 9)$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 33 (d1-033)

Difficulty: Foundation / Marks: 6

The curve C has equation $y = 2x^2 - x$.

(a) Differentiate from first principles to find $\frac{dy}{dx}$.

(b) Find the gradient of C at the point $A(2, 6)$.

(c) Find the equation of the tangent to C at A .

Worked solution.

Step 1: (a) Find $f(x+h) - f(x)$.

$$[2(x+h)^2 - (x+h)] - [2x^2 - x] = 4xh + 2h^2 - h$$

Expand and cancel $2x^2$ and $-x$ terms.

Step 2: Divide by h and take the limit.

$$4x + 2h - 1 \xrightarrow{h \rightarrow 0} 4x - 1$$

$$\frac{dy}{dx} = 4x - 1.$$

Step 3: (b) Substitute $x = 2$.

$$4(2) - 1 = 7$$

Gradient at A is 7.

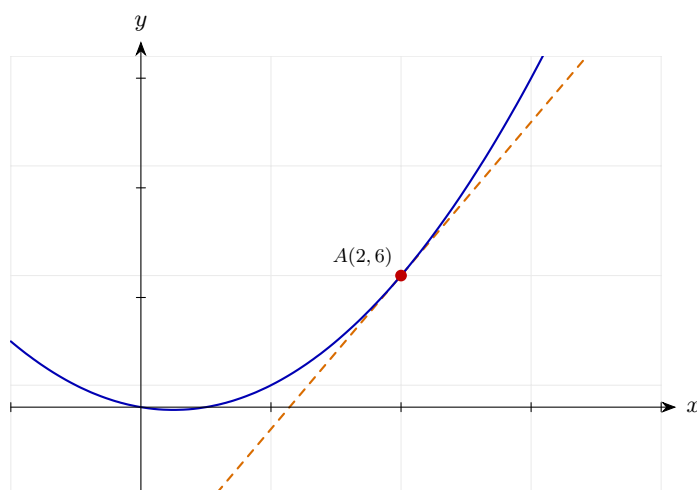
Step 4: (c) Equation of tangent using $y - 6 = 7(x - 2)$.

$$y = 7x - 14 + 6 = 7x - 8$$

Substitute $m = 7$ and point $(2, 6)$.

Final answer: (a) $\frac{dy}{dx} = 4x - 1$ (b) Gradient = 7 (c) $y = 7x - 8$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 34 (d1-034)

Difficulty: Foundation / Marks: 7

The curve C has equation $y = x^3 - 2$.

(a) By differentiating from first principles, show that $\frac{dy}{dx} = 3x^2$.

(b) Find the two points on C where the gradient is 12.

(c) Find the gradient of C at the origin.

Worked solution.

Step 1: (a) Differentiate $y = x^3 - 2$ from first principles.

$$\frac{(x+h)^3 - 2 - (x^3 - 2)}{h} = \frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2 \xrightarrow{h \rightarrow 0} 3x^2 \checkmark$$

The -2 terms cancel; the rest follows from expanding $(x+h)^3$.

Step 2: (b) Set $3x^2 = 12$.

$$x^2 = 4 \Rightarrow x = \pm 2$$

Two values of x .

Step 3: Find corresponding y -values.

$$x = 2 : y = 8 - 2 = 6 \quad x = -2 : y = -8 - 2 = -10$$

Substitute each x into $y = x^3 - 2$.

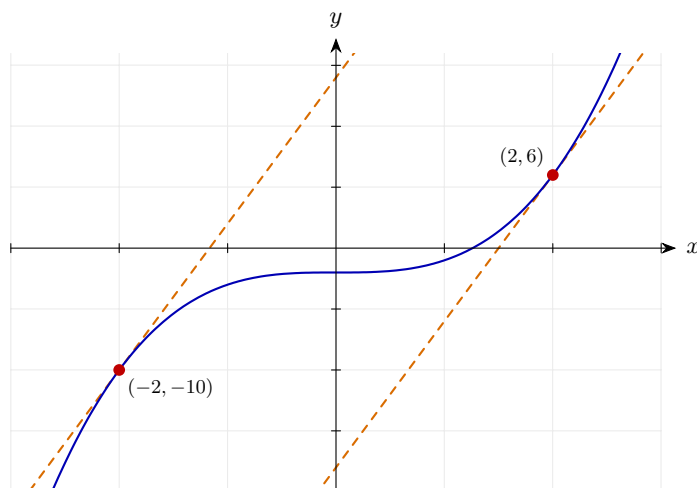
Step 4: (c) Gradient at origin ($x = 0$).

$$\left. \frac{dy}{dx} \right|_{x=0} = 3(0)^2 = 0$$

The curve has a stationary point at the origin.

Final answer: (b) $(2, 6)$ and $(-2, -10)$ (c) Gradient at origin is 0.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 35 (d1-035)

Difficulty: Foundation / Marks: 6

(a) Differentiate $f(x) = x^2$ from first principles to show that $f'(x) = 2x$.

(b) The tangent to $y = x^2$ at the point $P(a, a^2)$ has equation $y = 2ax - a^2$. Verify this using the result from part (a).

Worked solution.

Step 1: (a) From first principles (standard).

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x \checkmark$$

Standard first-principles derivation.

Step 2: (b) At $x = a$, the gradient is $f'(a) = 2a$.

$$\text{Gradient} = 2a$$

Substitute $x = a$ into the derivative.

Step 3: Use $y - y_1 = m(x - x_1)$ with $m = 2a$ and point (a, a^2) .

$$y - a^2 = 2a(x - a)$$

Point-gradient form of the line.

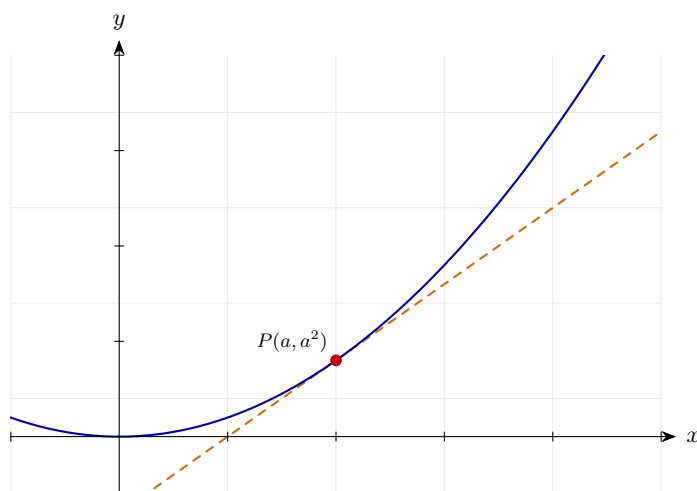
Step 4: Expand and simplify.

$$y = 2ax - 2a^2 + a^2 = 2ax - a^2 \checkmark$$

This matches the given tangent equation, confirming the result.

Final answer: Verified: tangent at $P(a, a^2)$ is $y = 2ax - a^2$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 36 (d1-036)

Difficulty: Standard / Marks: 4

The curve C has equation $y = x^2 + x$. Find the gradient of the chord from the point where $x = 1$ to the point where:

- (i) $x = 2$ (ii) $x = 1.1$ (iii) $x = 1.01$

Hence suggest the gradient of C at $x = 1$.

Worked solution.

Step 1: Let $f(x) = x^2 + x$. Then $f(1) = 2$.

$$\text{Chord gradient} = \frac{f(a) - 2}{a - 1}$$

Apply the chord-gradient formula.

Step 2: (i) $a = 2$, $f(2) = 6$.

$$\frac{6 - 2}{2 - 1} = 4$$

Chord gradient from $x = 1$ to $x = 2$.

Step 3: (ii) $a = 1.1$, $f(1.1) = 2.31$.

$$\frac{2.31 - 2}{1.1 - 1} = \frac{0.31}{0.1} = 3.1$$

Chord gradient closer to $x = 1$.

Step 4: (iii) $a = 1.01$, $f(1.01) = 2.0301$.

$$\frac{2.0301 - 2}{0.01} = 3.01$$

Chord gradient very close to $x = 1$.

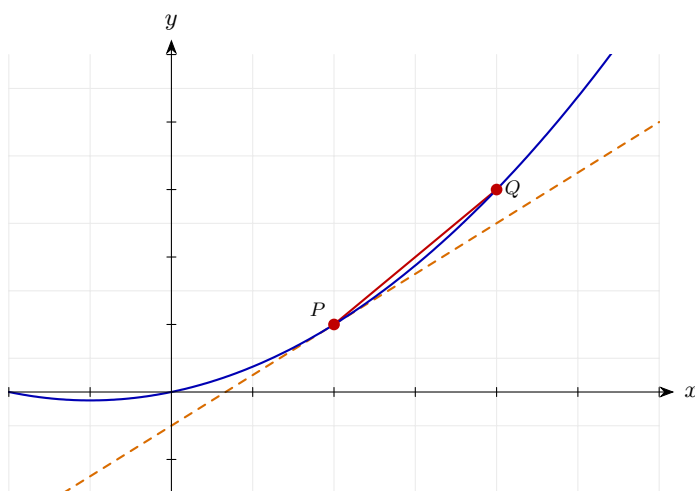
Step 5: The chord gradients approach 3.

Gradient at $x = 1 \approx 3$

Since $f'(x) = 2x + 1$, $f'(1) = 3$.

Final answer: (i) 4 (ii) 3.1 (iii) 3.01 Gradient at $x = 1$ is 3.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 37 (d1-037)

Difficulty: Standard / Marks: 4

The curve C is given by $y = x^3 - x$. Find the gradient of the chord from the point $(1, 0)$ to the point where:

- (i) $x = 1.5$ (ii) $x = 1.1$ (iii) $x = 1.01$

What does this suggest about the gradient of C at $x = 1$?

Worked solution.

Step 1: Let $f(x) = x^3 - x$, so $f(1) = 0$.

$$\text{Chord gradient} = \frac{f(a) - 0}{a - 1}$$

Chord from $(1, 0)$ to $(a, f(a))$.

Step 2: (i) $a = 1.5$, $f(1.5) = 1.875$.

$$\frac{1.875}{0.5} = 3.75$$

Chord gradient.

Step 3: (ii) $a = 1.1$, $f(1.1) = 0.231$.

$$\frac{0.231}{0.1} = 2.31$$

Chord gradient.

Step 4: (iii) $a = 1.01$, $f(1.01) = 0.020301$.

$$\frac{0.020301}{0.01} = 2.0301$$

Chord gradient.

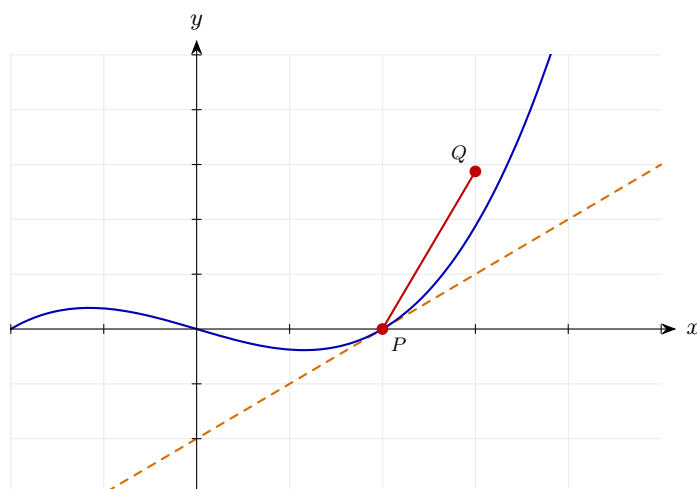
Step 5: The chord gradients approach 2.

Gradient at $x = 1 \approx 2$

$f'(x) = 3x^2 - 1$, so $f'(1) = 2$.

Final answer: (i) 3.75 (ii) 2.31 (iii) 2.0301 Gradient at $x = 1$ is 2.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 38 (d1-038)

Difficulty: Standard / Marks: 4

For the curve $y = 2x^2$, find the gradient of the chord joining $x = -1$ to:

(i) $x = 0$ (ii) $x = -0.5$ (iii) $x = -0.9$

Deduce the gradient of the curve at $x = -1$.

Worked solution.

Step 1: Let $f(x) = 2x^2$, $f(-1) = 2$.

$$\text{Chord gradient} = \frac{f(a) - 2}{a - (-1)} = \frac{2a^2 - 2}{a + 1}$$

Chord-gradient formula.

Step 2: (i) $a = 0$.

$$\frac{0 - 2}{0 + 1} = -2$$

Chord gradient.

Step 3: (ii) $a = -0.5$, $f(-0.5) = 0.5$.

$$\frac{0.5 - 2}{0.5} = -3$$

Chord gradient.

Step 4: (iii) $a = -0.9$, $f(-0.9) = 1.62$.

$$\frac{1.62 - 2}{0.1} = -3.8$$

Chord gradient.

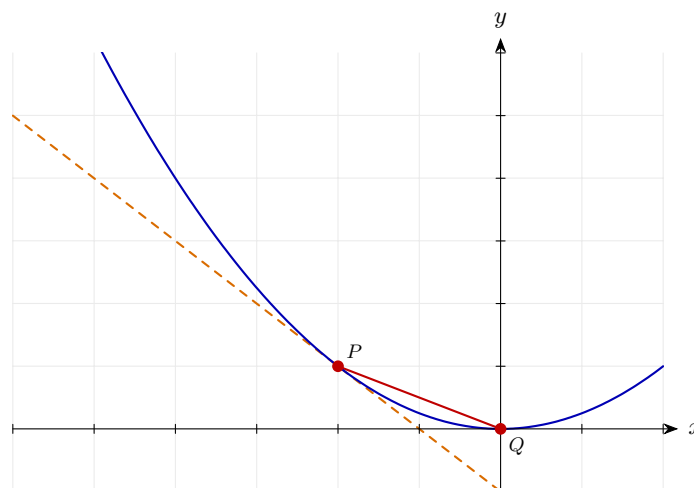
Step 5: The chord gradients approach -4 .

$$f'(x) = 4x \Rightarrow f'(-1) = -4$$

Consistent with differentiation.

Final answer: (i) -2 (ii) -3 (iii) -3.8 Gradient at $x = -1$ is -4 .

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 39 (d1-039)

Difficulty: Standard / Marks: 4

A student estimates the gradient of $y = x^2$ at $x = 4$ using chords to $x = 4 + h$ for $h = 1, 0.1, 0.01$. Tabulate the three chord gradients and state what the gradient at $x = 4$ appears to be.

Worked solution.

$$\text{Step 1: Chord gradient} = \frac{(4+h)^2 - 16}{h} = 8 + h.$$

$$\frac{(4+h)^2 - 16}{h} = \frac{8h + h^2}{h} = 8 + h$$

Simplify using algebra.

$$\text{Step 2: } h = 1 \Rightarrow \text{gradient} = 9.$$

$$8 + 1 = 9$$

Substitute $h = 1$.

$$\text{Step 3: } h = 0.1 \Rightarrow \text{gradient} = 8.1.$$

$$8 + 0.1 = 8.1$$

Substitute $h = 0.1$.

$$\text{Step 4: } h = 0.01 \Rightarrow \text{gradient} = 8.01.$$

$$8 + 0.01 = 8.01$$

Substitute $h = 0.01$.

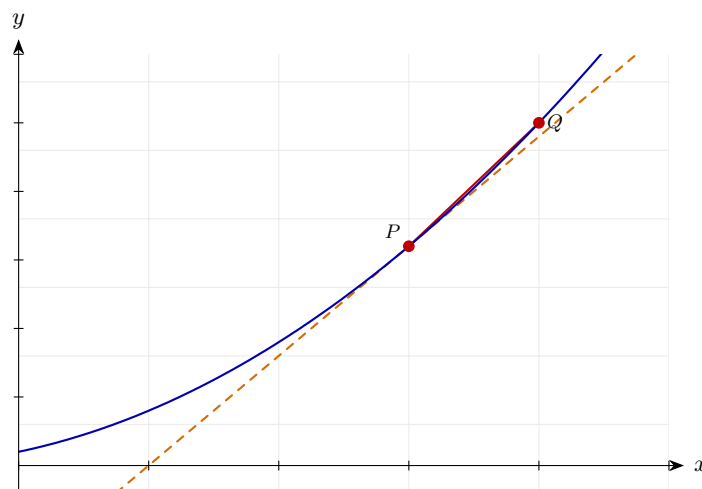
$$\text{Step 5: As } h \rightarrow 0, \text{ chord gradient} \rightarrow 8.$$

$$\text{Gradient at } x = 4 = 8$$

The limit is 8. The diagram shows how the chords flatten onto the tangent at $(4, 16)$ as h shrinks.

Final answer: Chord gradients: 9, 8.1, 8.01. Gradient of curve at $x = 4$ is 8.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 40 (d1-040)

Difficulty: Standard / Marks: 5

The curve C has equation $y = 3 - x^2$. Find the gradient of the chord joining $(2, -1)$ to the point where $x = 2 + h$. Simplify your answer and deduce the gradient of C at $x = 2$ by letting $h \rightarrow 0$.

Worked solution.

Step 1: Compute $f(2 + h)$ where $f(x) = 3 - x^2$.

$$f(2 + h) = 3 - (2 + h)^2 = 3 - 4 - 4h - h^2 = -1 - 4h - h^2$$

Expand $(2 + h)^2$.

Step 2: Chord gradient $= \frac{f(2 + h) - f(2)}{h}$.

$$\frac{(-1 - 4h - h^2) - (-1)}{h} = \frac{-4h - h^2}{h}$$

Subtract $f(2) = -1$.

Step 3: Simplify.

$$= -4 - h$$

Divide every term by h .

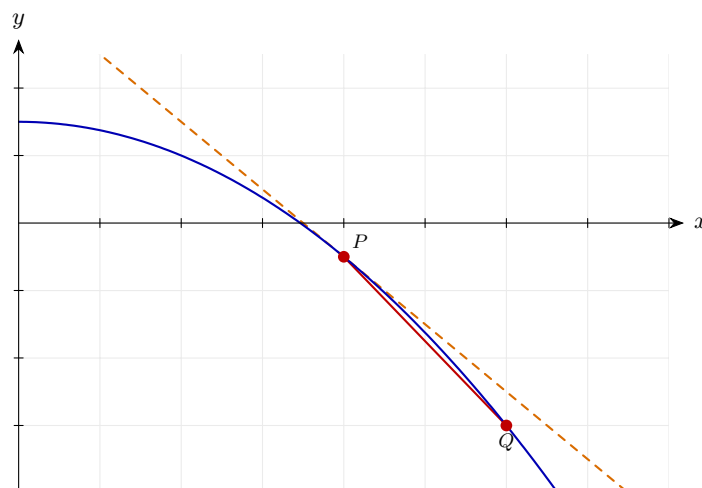
Step 4: Let $h \rightarrow 0$.

$$\lim_{h \rightarrow 0} (-4 - h) = -4$$

The chord gradient tends to -4 .

Final answer: Chord gradient $= -4 - h$. Gradient of C at $x = 2$ is -4 .

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 41 (d1-041)

Difficulty: Standard / Marks: 4

Differentiate $f(x) = 3x^2$ from first principles.

Worked solution.

Step 1: Write down the definition of the derivative.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Standard first-principles definition.

Step 2: Compute $f(x+h)$.

$$f(x+h) = 3(x+h)^2 = 3x^2 + 6xh + 3h^2$$

Expand $(x+h)^2$ and multiply by 3.

Step 3: Form the difference quotient.

$$\frac{3x^2 + 6xh + 3h^2 - 3x^2}{h} = \frac{6xh + 3h^2}{h} = 6x + 3h$$

Subtract, then divide by h .

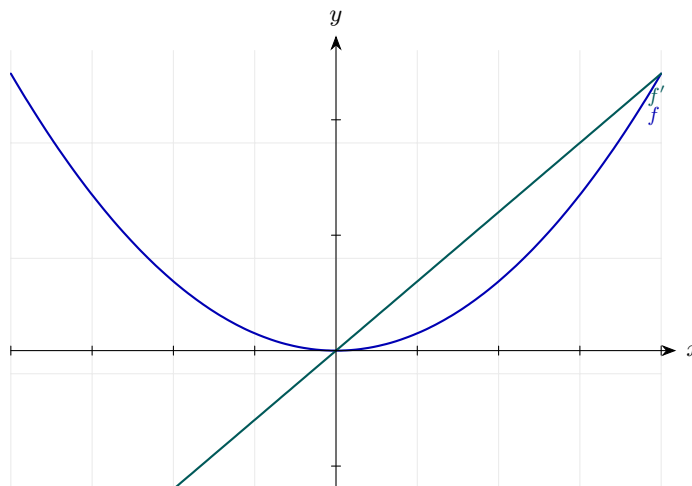
Step 4: Take the limit.

$$f'(x) = \lim_{h \rightarrow 0} (6x + 3h) = 6x$$

The $3h$ term vanishes.

Final answer: $f'(x) = 6x$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 42 (d1-042)

Difficulty: Standard / Marks: 4

Differentiate $f(x) = x^2 - 4x$ from first principles.

Worked solution.

Step 1: Definition of derivative.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Standard definition.

Step 2: Compute $f(x+h)$.

$$f(x+h) = (x+h)^2 - 4(x+h) = x^2 + 2xh + h^2 - 4x - 4h$$

Expand brackets.

Step 3: Form difference quotient.

$$\frac{2xh + h^2 - 4h}{h} = 2x + h - 4$$

x^2 and $-4x$ terms cancel.

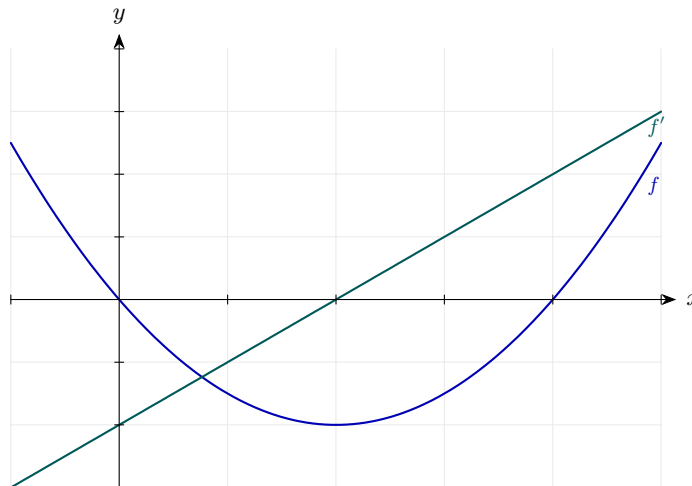
Step 4: Take the limit as $h \rightarrow 0$.

$$f'(x) = 2x - 4$$

The h term vanishes.

Final answer: $f'(x) = 2x - 4$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 43 (d1-043)

Difficulty: Standard / Marks: 4

Use first principles to differentiate $f(x) = x^2 + 3x - 1$.

Worked solution.

Step 1: Compute $f(x+h)$.

$$f(x+h) = (x+h)^2 + 3(x+h) - 1 = x^2 + 2xh + h^2 + 3x + 3h - 1$$

Expand all terms.

Step 2: Compute $f(x+h) - f(x)$.

$$2xh + h^2 + 3h$$

Constant and x terms cancel.

Step 3: Divide by h .

$$\frac{2xh + h^2 + 3h}{h} = 2x + h + 3$$

Simplify.

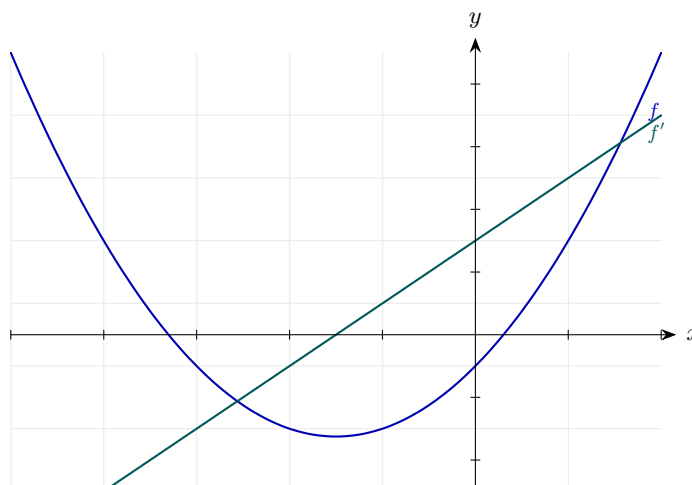
Step 4: Let $h \rightarrow 0$.

$$f'(x) = 2x + 3$$

The -1 constant has no effect on the derivative.

Final answer: $f'(x) = 2x + 3$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 44 (d1-044)

Difficulty: Standard / Marks: 5

Differentiate $f(x) = 2x^3$ from first principles.

Worked solution.

Step 1: Expand $(x+h)^3$.

$$(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$$

Binomial expansion.

Step 2: Compute $f(x+h) - f(x)$.

$$2(x^3 + 3x^2h + 3xh^2 + h^3) - 2x^3 = 6x^2h + 6xh^2 + 2h^3$$

Subtract and collect terms.

Step 3: Divide by h .

$$\frac{6x^2h + 6xh^2 + 2h^3}{h} = 6x^2 + 6xh + 2h^2$$

Cancel one factor of h .

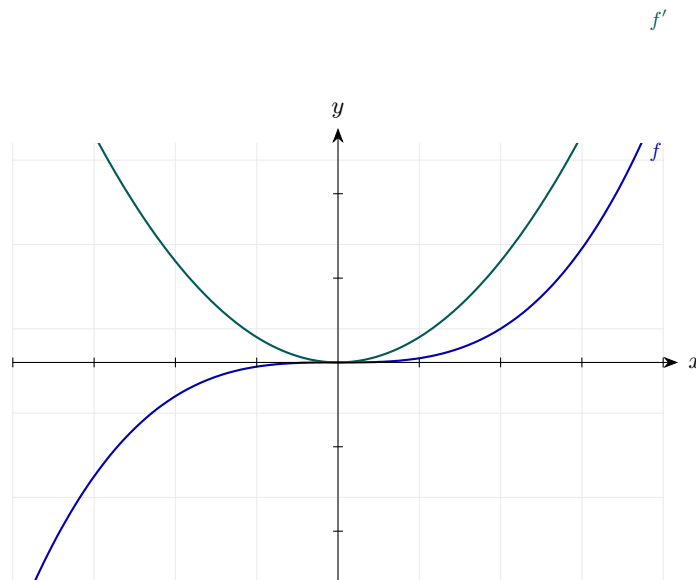
Step 4: Take the limit.

$$f'(x) = \lim_{h \rightarrow 0} (6x^2 + 6xh + 2h^2) = 6x^2$$

All h terms vanish.

Final answer: $f'(x) = 6x^2$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 45 (d1-045)

Difficulty: Standard / Marks: 5

Differentiate $f(x) = x^3 + 2x$ **from first principles.**

Worked solution.

Step 1: Compute $f(x + h)$.

$$(x + h)^3 + 2(x + h) = x^3 + 3x^2h + 3xh^2 + h^3 + 2x + 2h$$

Expand cube and linear term.

Step 2: Subtract $f(x) = x^3 + 2x$.

$$3x^2h + 3xh^2 + h^3 + 2h$$

x^3 and $2x$ cancel.

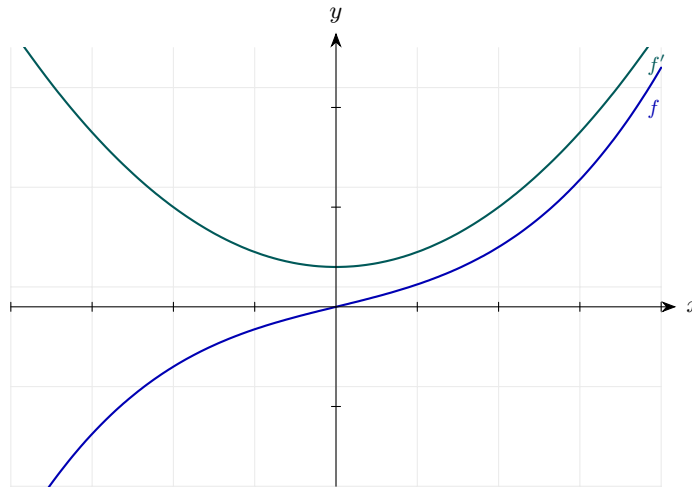
Step 3: Divide by h and take the limit.

$$f'(x) = \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2 + 2) = 3x^2 + 2$$

All h terms vanish.

Final answer: $f'(x) = 3x^2 + 2$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 46 (d1-046)

Difficulty: Standard / Marks: 4

Use first principles to show that the derivative of $f(x) = 5x^2 - 3$ is $10x$.

Worked solution.

Step 1: Compute $f(x + h)$.

$$f(x + h) = 5(x + h)^2 - 3 = 5x^2 + 10xh + 5h^2 - 3$$

Expand and simplify.

Step 2: Form the difference quotient.

$$\frac{(5x^2 + 10xh + 5h^2 - 3) - (5x^2 - 3)}{h} = \frac{10xh + 5h^2}{h} = 10x + 5h$$

Constants cancel.

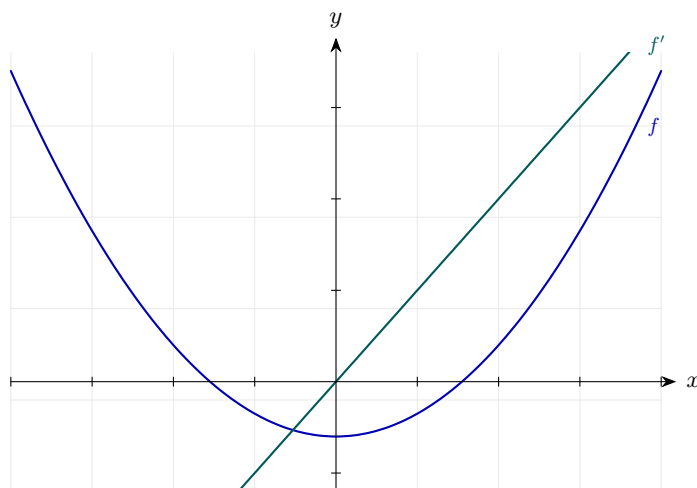
Step 3: Take the limit.

$$f'(x) = \lim_{h \rightarrow 0} (10x + 5h) = 10x \checkmark$$

The $5h$ term vanishes.

Final answer: Shown: $f'(x) = 10x$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 47 (d1-047)

Difficulty: Standard / Marks: 5

Use first principles to find the gradient of the curve $y = x^2 + 2x$ at the point where $x = 3$.

Worked solution.

Step 1: Let $f(x) = x^2 + 2x$. Form $f(3+h) - f(3)$.

$$f(3+h) = (3+h)^2 + 2(3+h) = 9 + 6h + h^2 + 6 + 2h = 15 + 8h + h^2$$

$$f(3) = 15.$$

Step 2: Subtract $f(3)$ and divide by h .

$$\frac{8h + h^2}{h} = 8 + h$$

Simplify the difference quotient.

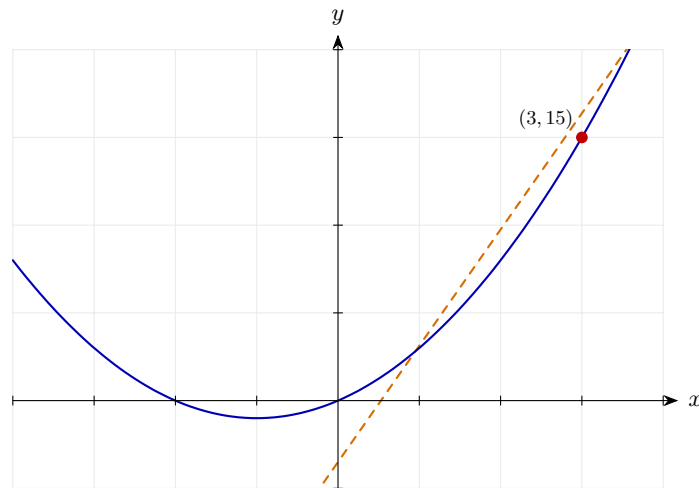
Step 3: Take the limit.

$$f'(3) = \lim_{h \rightarrow 0} (8 + h) = 8$$

The gradient at $x = 3$ is 8.

Final answer: Gradient at $x = 3$ is 8.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 48 (d1-048)

Difficulty: Standard / Marks: 5

Find, from first principles, the gradient of the curve $y = x^3$ at the point where $x = -2$.

Worked solution.

Step 1: Let $f(x) = x^3$. Compute $f(-2 + h)$.

$$(-2 + h)^3 = -8 + 12h - 6h^2 + h^3$$

Expand the cube.

Step 2: Subtract $f(-2) = -8$.

$$12h - 6h^2 + h^3$$

The -8 cancels.

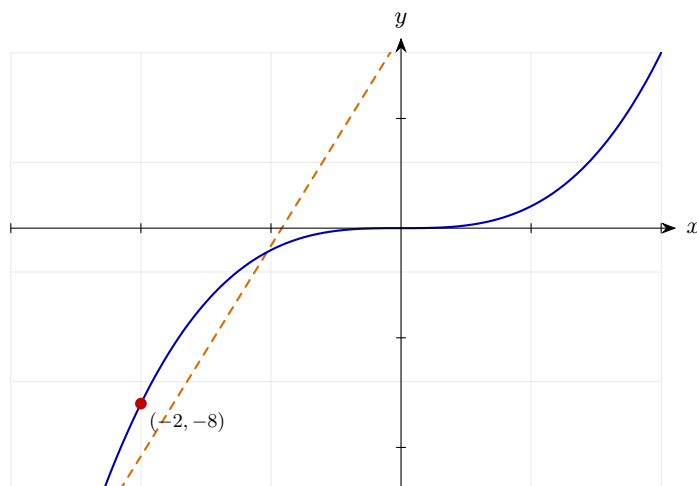
Step 3: Divide by h and take the limit.

$$f'(-2) = \lim_{h \rightarrow 0} (12 - 6h + h^2) = 12$$

The h terms vanish.

Final answer: Gradient at $x = -2$ is 12.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 49 (d1-049)

Difficulty: Standard / Marks: 5

Using first principles, find the gradient of the curve $y = 4x^2 - x$ at $x = 1$.

Worked solution.

Step 1: Let $f(x) = 4x^2 - x$. Then $f(1) = 3$.

$$f(1+h) = 4(1+h)^2 - (1+h) = 4 + 8h + 4h^2 - 1 - h = 3 + 7h + 4h^2$$

Expand and simplify.

Step 2: Form the difference quotient.

$$\frac{(3 + 7h + 4h^2) - 3}{h} = 7 + 4h$$

Cancel and divide.

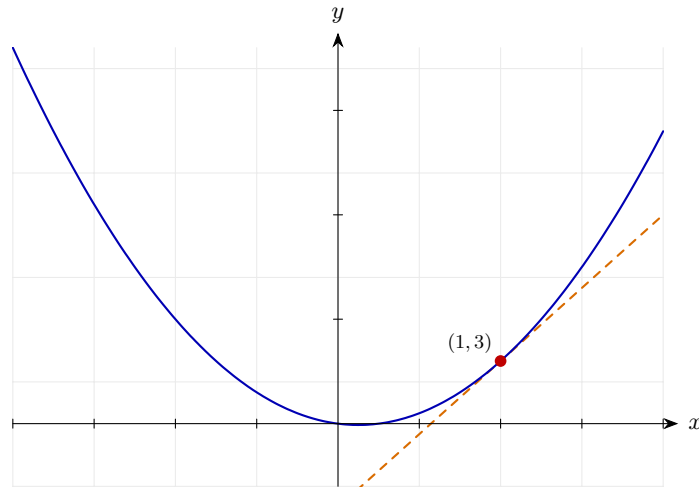
Step 3: Take the limit.

$$f'(1) = \lim_{h \rightarrow 0} (7 + 4h) = 7$$

Limit as $h \rightarrow 0$.

Final answer: Gradient at $x = 1$ is 7.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 50 (d1-050)

Difficulty: Standard / Marks: 5

A curve has equation $y = x^2 - 6x + 5$. Use first principles to show that the gradient of the curve at $x = 4$ is 2.

Worked solution.

Step 1: Let $f(x) = x^2 - 6x + 5$. Compute $f(4)$.

$$f(4) = 16 - 24 + 5 = -3$$

Substitute $x = 4$.

Step 2: Compute $f(4 + h)$.

$$(4 + h)^2 - 6(4 + h) + 5 = 16 + 8h + h^2 - 24 - 6h + 5 = -3 + 2h + h^2$$

Expand and simplify.

Step 3: Difference quotient.

$$\frac{(-3 + 2h + h^2) - (-3)}{h} = 2 + h$$

Cancel and divide.

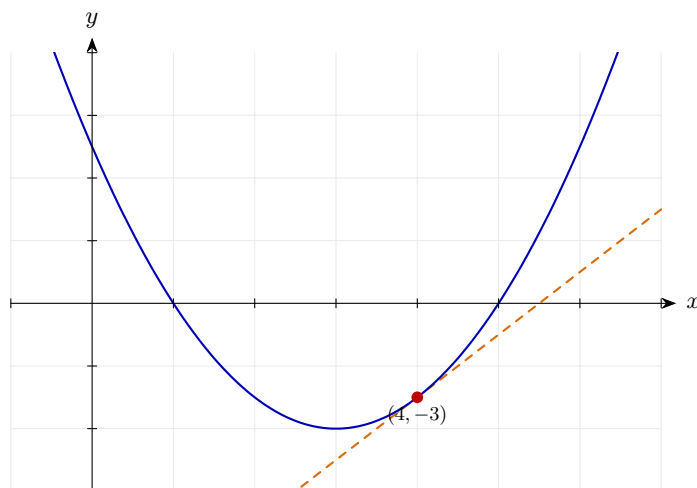
Step 4: Take the limit.

$$f'(4) = \lim_{h \rightarrow 0} (2 + h) = 2 \checkmark$$

As required.

Final answer: Shown: gradient at $x = 4$ is 2.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 51 (d1-051)

Difficulty: Standard / Marks: 5

The curve C has equation $y = 2x^2 + 5x$. Find the gradient of C at the point where $x = -1$, using first principles.

Worked solution.

Step 1: Let $f(x) = 2x^2 + 5x$. Then $f(-1) = 2 - 5 = -3$.

$$f(-1 + h) = 2(-1 + h)^2 + 5(-1 + h)$$

Compute $f(-1 + h)$.

Step 2: Expand.

$$2(1 - 2h + h^2) + (-5 + 5h) = 2 - 4h + 2h^2 - 5 + 5h = -3 + h + 2h^2$$

Collect like terms.

Step 3: Form the difference quotient.

$$\frac{(-3 + h + 2h^2) - (-3)}{h} = 1 + 2h$$

Cancel -3 , divide by h .

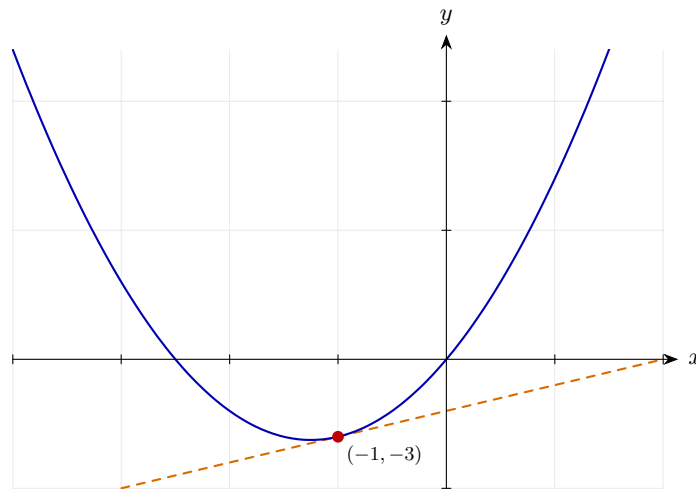
Step 4: Take the limit.

$$f'(-1) = \lim_{h \rightarrow 0} (1 + 2h) = 1$$

Limit as $h \rightarrow 0$.

Final answer: Gradient at $x = -1$ is 1.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 52 (d1-052)

Difficulty: Standard / Marks: 5

Find, from first principles, the gradient of $y = x^3 + x$ at the point where $x = 2$.

Worked solution.

Step 1: Let $f(x) = x^3 + x$. Then $f(2) = 10$.

$$f(2+h) = (2+h)^3 + (2+h) = 8 + 12h + 6h^2 + h^3 + 2 + h$$

Expand cube and linear term.

Step 2: Simplify.

$$= 10 + 13h + 6h^2 + h^3$$

Combine constant terms.

Step 3: Difference quotient.

$$\frac{13h + 6h^2 + h^3}{h} = 13 + 6h + h^2$$

Cancel and divide.

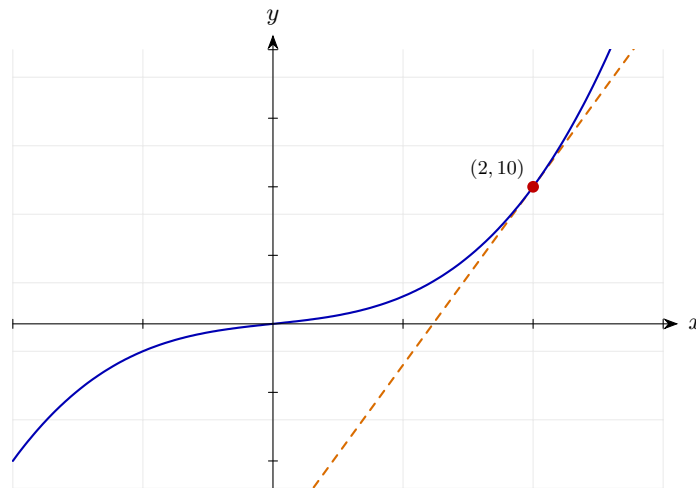
Step 4: Take the limit.

$$f'(2) = \lim_{h \rightarrow 0} (13 + 6h + h^2) = 13$$

Limit as $h \rightarrow 0$.

Final answer: Gradient at $x = 2$ is 13.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 53 (d1-053)

Difficulty: Standard / Marks: 5

A curve has equation $y = x^2 + 4x - 7$. (a) Find $\frac{dy}{dx}$ from first principles. (b) Hence find the gradient of the curve at the point where $x = -3$.

Worked solution.

Step 1: (a) Compute $f(x+h) - f(x)$.

$$[(x+h)^2 + 4(x+h) - 7] - [x^2 + 4x - 7] = 2xh + h^2 + 4h$$

Expand and subtract.

Step 2: Divide by h and take the limit.

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} (2x + h + 4) = 2x + 4$$

The gradient function.

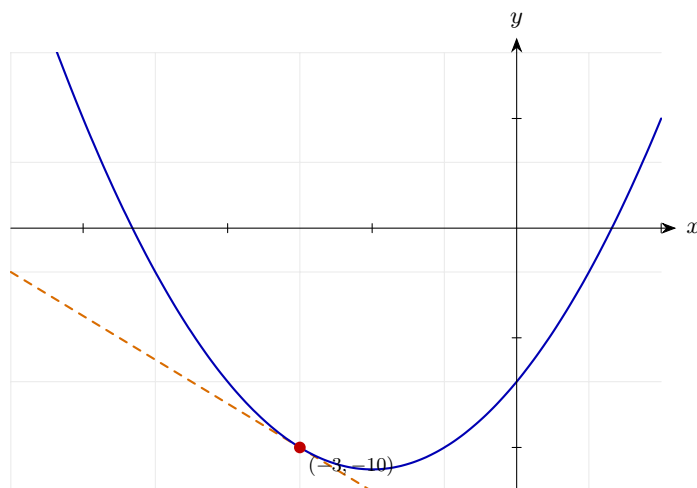
Step 3: (b) Substitute $x = -3$.

$$2(-3) + 4 = -2$$

Evaluate the derivative.

Final answer: (a) $\frac{dy}{dx} = 2x + 4$ (b) Gradient at $x = -3$ is -2 .

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 54 (d1-054)

Difficulty: Standard / Marks: 6

The curve C has equation $y = x^2 - 5x$. (a) Use first principles to find the gradient function. (b) Find the value of x at which the gradient of C is zero.

Worked solution.

Step 1: (a) Difference quotient.

$$\frac{(x+h)^2 - 5(x+h) - (x^2 - 5x)}{h} = \frac{2xh + h^2 - 5h}{h} = 2x + h - 5$$

Expand, subtract, divide.

Step 2: Take the limit.

$$\frac{dy}{dx} = 2x - 5$$

Gradient function.

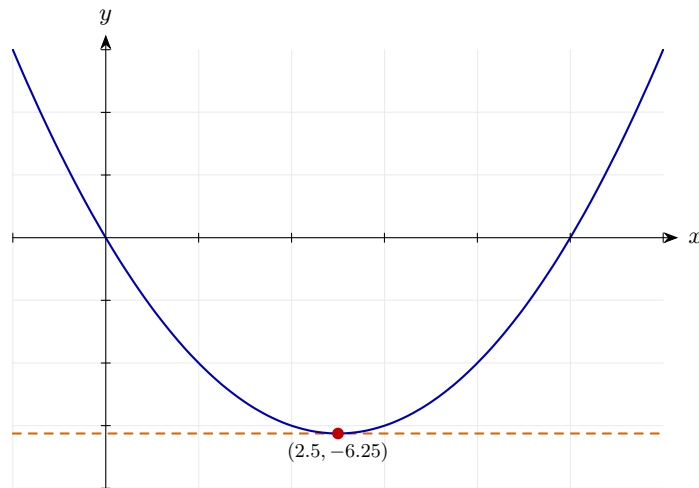
Step 3: (b) Set gradient = 0.

$$2x - 5 = 0 \Rightarrow x = \frac{5}{2}$$

Solve for x . At this x the tangent is horizontal — the curve has its minimum.

Final answer: (a) $\frac{dy}{dx} = 2x - 5$ (b) $x = \frac{5}{2}$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 55 (d1-055)

Difficulty: Standard / Marks: 6

A curve has equation $y = 3x^2 - 12x + 5$. Using first principles, find the value of x at which the tangent to the curve is horizontal.

Worked solution.

Step 1: Compute the difference quotient.

$$\frac{3(x+h)^2 - 12(x+h) + 5 - (3x^2 - 12x + 5)}{h} = \frac{6xh + 3h^2 - 12h}{h}$$

Expand and simplify.

Step 2: Simplify and take the limit.

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} (6x + 3h - 12) = 6x - 12$$

Gradient function.

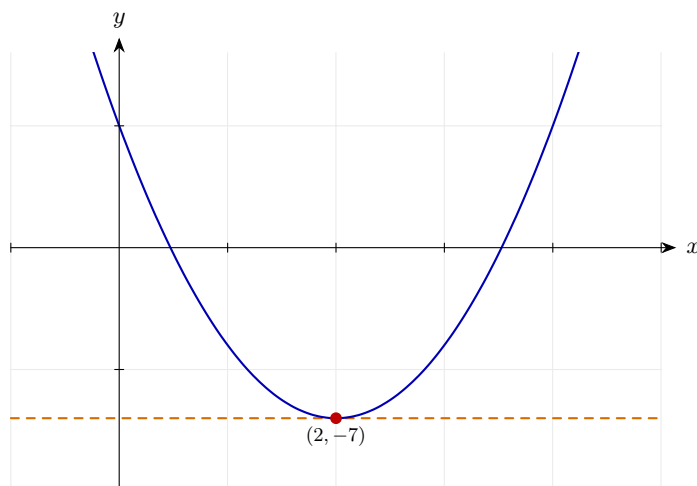
Step 3: Tangent horizontal when $\frac{dy}{dx} = 0$.

$$6x - 12 = 0 \Rightarrow x = 2$$

Solve for x .

Final answer: Horizontal tangent at $x = 2$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 56 (d1-056)

Difficulty: Standard / Marks: 6

The curve C is given by $y = x^3 - 3x$. (a) Find $\frac{dy}{dx}$ from first principles. (b) Find the gradient of C at the points where $x = 0$ and $x = 1$.

Worked solution.

Step 1: (a) Compute $f(x+h) - f(x)$.

$$(x+h)^3 - 3(x+h) - x^3 + 3x = 3x^2h + 3xh^2 + h^3 - 3h$$

Expand the cube.

Step 2: Divide by h and take the limit.

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2 - 3) = 3x^2 - 3$$

Gradient function.

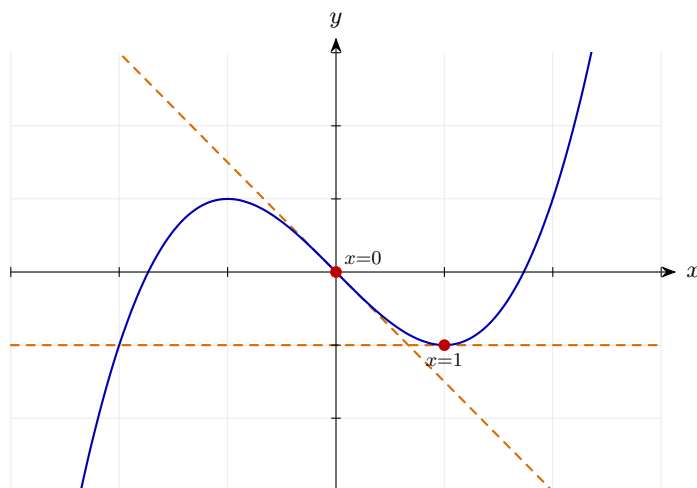
Step 3: (b) Substitute.

$$\left. \frac{dy}{dx} \right|_{x=0} = -3, \quad \left. \frac{dy}{dx} \right|_{x=1} = 0$$

Evaluate at both values.

Final answer: (a) $\frac{dy}{dx} = 3x^2 - 3$ (b) Gradients are -3 and 0 .

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 57 (d1-057)

Difficulty: Standard / Marks: 5

The curve C has equation $y = x^2 + 2x - 3$. Find the coordinates of the point on C at which the gradient is equal to 6.

Worked solution.

Step 1: Differentiate (by first principles or directly).

$$\frac{dy}{dx} = 2x + 2$$

Gradient function for a quadratic.

Step 2: Set gradient = 6.

$$2x + 2 = 6 \Rightarrow x = 2$$

Solve for x .

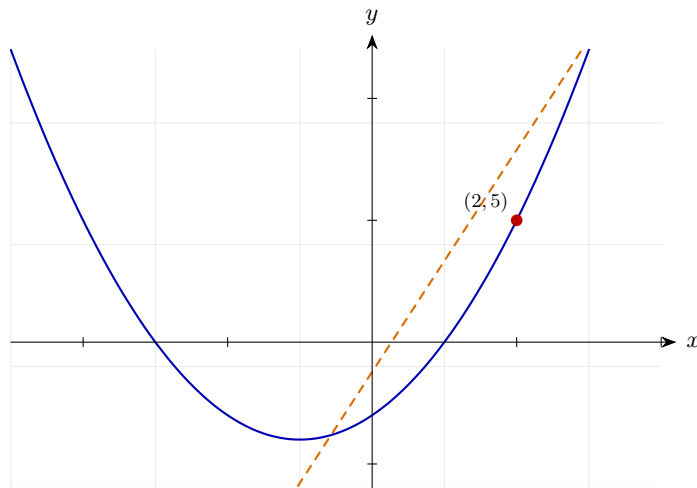
Step 3: Find y .

$$y = 4 + 4 - 3 = 5$$

Substitute $x = 2$.

Final answer: Point is $(2, 5)$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 58 (d1-058)

Difficulty: Standard / Marks: 4

Find the coordinates of the point on the curve $y = x^2 - 4x + 1$ where the gradient is -2 .

Worked solution.

Step 1: Differentiate.

$$\frac{dy}{dx} = 2x - 4$$

Gradient function.

Step 2: Set gradient equal to -2 .

$$2x - 4 = -2 \Rightarrow x = 1$$

Solve for x .

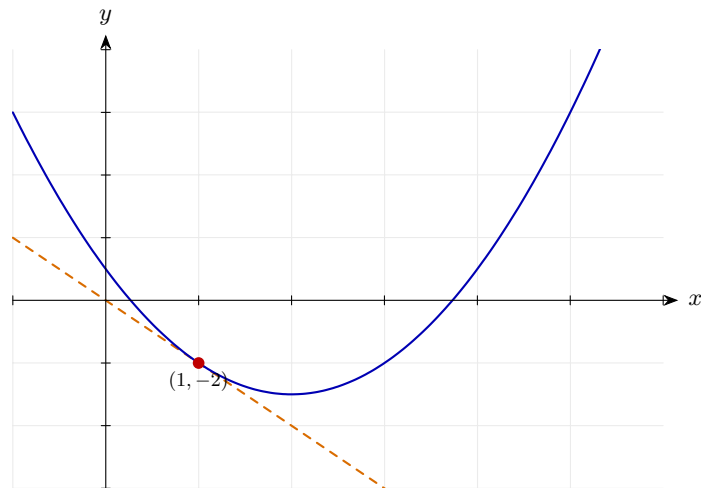
Step 3: Find y .

$$y = 1 - 4 + 1 = -2$$

Substitute $x = 1$.

Final answer: Point is $(1, -2)$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 59 (d1-059)

Difficulty: Standard / Marks: 4

The displacement s metres of a particle at time t seconds is modelled by $s = t^2 + 3t$. (a) Find the velocity $\frac{ds}{dt}$. (b) Find the velocity when $t = 4$.

Worked solution.

Step 1: (a) Differentiate s with respect to t .

$$\frac{ds}{dt} = 2t + 3$$

Gradient function — the velocity.

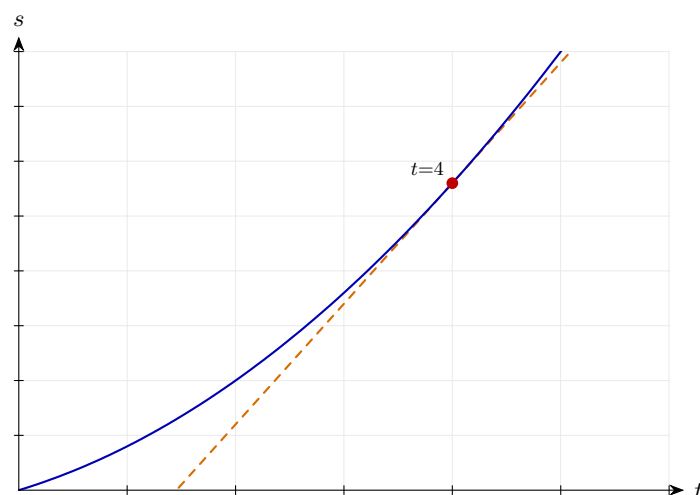
Step 2: (b) Substitute $t = 4$.

$$2(4) + 3 = 11$$

Units: m/s.

Final answer: (a) $\frac{ds}{dt} = 2t + 3$ (b) Velocity is 11 m/s.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 60 (d1-060)

Difficulty: Standard / Marks: 4

A ball is thrown so that its height h metres after t seconds is $h = 20t - 5t^2$. Find the rate at which the height is changing when $t = 1$.

Worked solution.

Step 1: Differentiate with respect to t .

$$\frac{dh}{dt} = 20 - 10t$$

Rate of change of height.

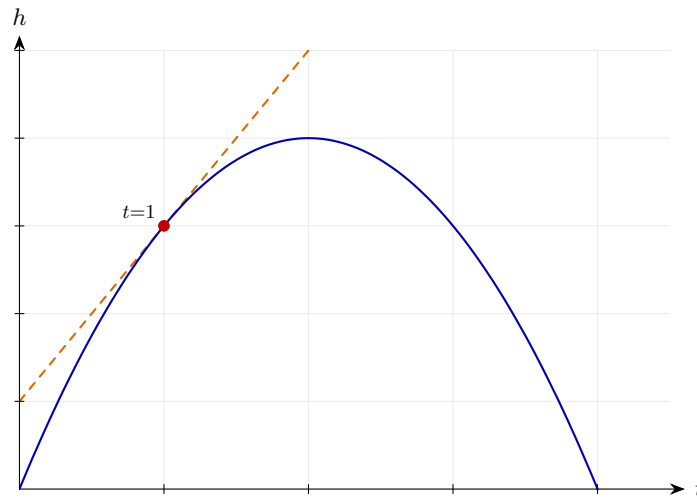
Step 2: Substitute $t = 1$.

$$20 - 10(1) = 10$$

The ball is rising at 10 m/s. The diagram shows the height-time curve and the tangent at $t = 1$, whose gradient 10 is the instantaneous velocity.

Final answer: Rate of change of height is 10 m/s (rising).

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 61 (d1-061)

Difficulty: Standard / Marks: 4

The volume V cm³ of a balloon at time t seconds is $V = 4t^2 + t$. Find $\frac{dV}{dt}$ and state the rate at which the volume is increasing when $t = 3$.

Worked solution.

Step 1: Differentiate.

$$\frac{dV}{dt} = 8t + 1$$

Rate of change of volume.

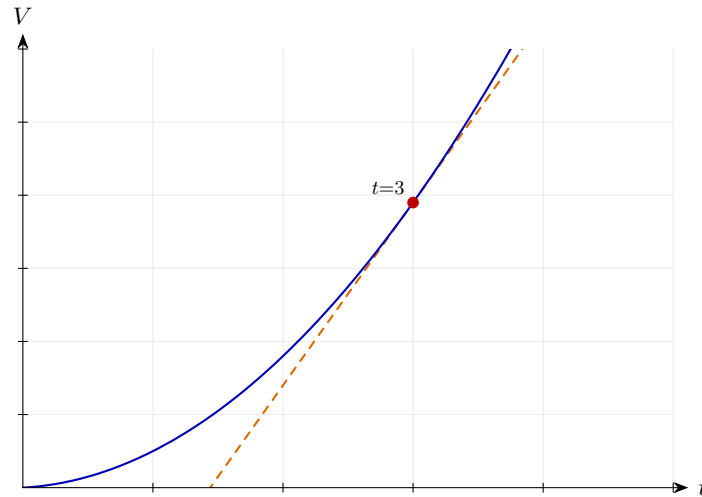
Step 2: Substitute $t = 3$.

$$8(3) + 1 = 25$$

Units: cm^3/s .

Final answer: $\frac{dV}{dt} = 8t + 1$; rate = $25 \text{ cm}^3/\text{s}$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 62 (d1-062)

Difficulty: Standard / Marks: 5

A company models its profit P (in £thousands) as a function of units sold x by $P = -x^2 + 40x - 100$. Find $\frac{dP}{dx}$ and determine the value of x at which profit is momentarily constant (the marginal profit is zero).

Worked solution.

Step 1: Differentiate.

$$\frac{dP}{dx} = -2x + 40$$

Marginal profit.

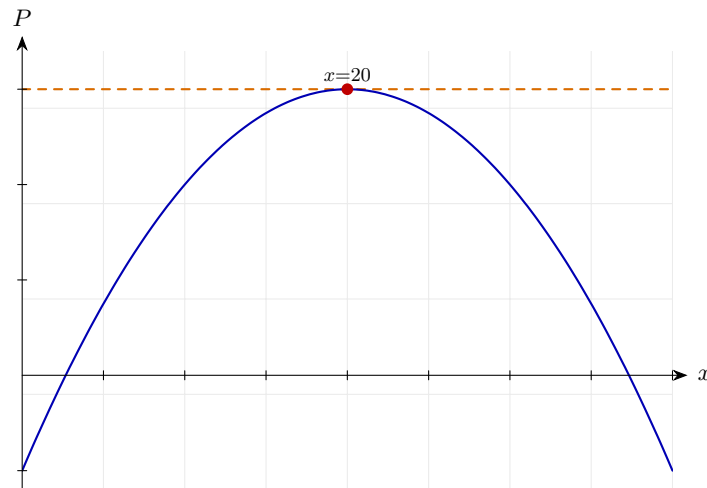
Step 2: Set equal to zero.

$$-2x + 40 = 0 \Rightarrow x = 20$$

Solve for x .

Final answer: $\frac{dP}{dx} = -2x + 40$; profit is momentarily constant at $x = 20$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 63 (d1-063)

Difficulty: Standard / Marks: 4

A cup of hot coffee cools so that its temperature T °C after t minutes is modelled by $T = 90 - 2t^2$. Find the rate of change of temperature when $t = 3$ and interpret your answer.

Worked solution.

Step 1: Differentiate.

$$\frac{dT}{dt} = -4t$$

Rate of change of temperature.

Step 2: Substitute $t = 3$.

$$-4(3) = -12$$

Negative means the coffee is cooling.

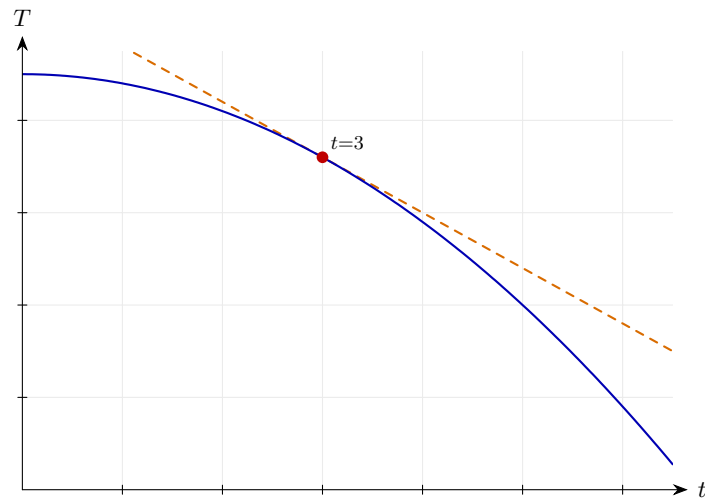
Step 3: Interpret.

Cooling at 12 °C per minute

The temperature is decreasing at 12 °C/min at this instant.

Final answer: Rate = -12 °C/min — the coffee is cooling at 12 °C/min.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 64 (d1-064)

Difficulty: Challenge / Marks: 6

Differentiate $f(x) = x^4$ from first principles.

Worked solution.

Step 1: Expand $(x + h)^4$.

$$(x + h)^4 = x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4$$

Binomial expansion.

Step 2: Form $f(x + h) - f(x)$.

$$4x^3h + 6x^2h^2 + 4xh^3 + h^4$$

x^4 cancels.

Step 3: Divide by h .

$$4x^3 + 6x^2h + 4xh^2 + h^3$$

Cancel one factor of h .

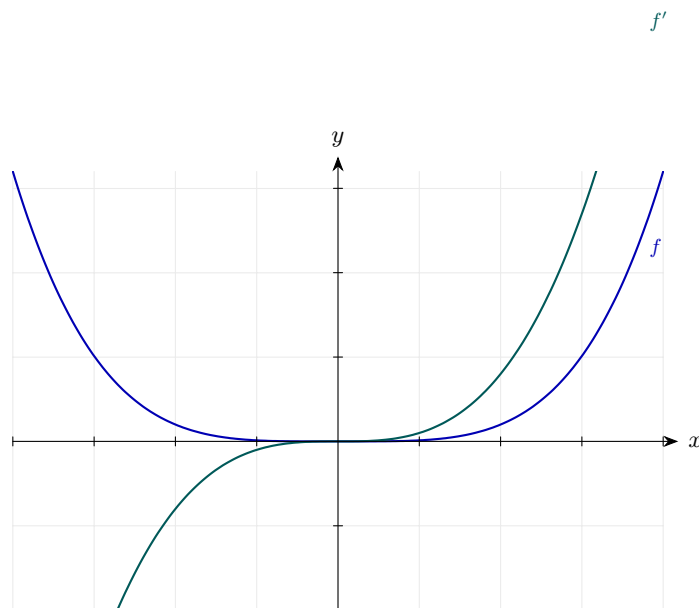
Step 4: Take the limit as $h \rightarrow 0$.

$$f'(x) = 4x^3$$

All h terms vanish.

Final answer: $f'(x) = 4x^3$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 65 (d1-065)

Difficulty: Challenge / Marks: 8

The curve C has equation $y = x^3 - 2x^2 + x$. (a) Use first principles to find $\frac{dy}{dx}$. (b) Hence find the values of x at which the tangent to C has gradient 1.

Worked solution.

Step 1: (a) Compute $f(x+h)$.

$$(x+h)^3 - 2(x+h)^2 + (x+h)$$

Expand term by term.

Step 2: Expand and collect.

$$= x^3 + 3x^2h + 3xh^2 + h^3 - 2x^2 - 4xh - 2h^2 + x + h$$

Binomial expansion of each bracket.

Step 3: Subtract $f(x) = x^3 - 2x^2 + x$.

$$3x^2h + 3xh^2 + h^3 - 4xh - 2h^2 + h$$

Cancel $f(x)$ terms.

Step 4: Divide by h and take the limit.

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2 - 4x - 2h + 1) = 3x^2 - 4x + 1$$

Gradient function.

Step 5: (b) Set gradient = 1.

$$3x^2 - 4x + 1 = 1 \Rightarrow 3x^2 - 4x = 0 \Rightarrow x(3x - 4) = 0$$

Factorise.

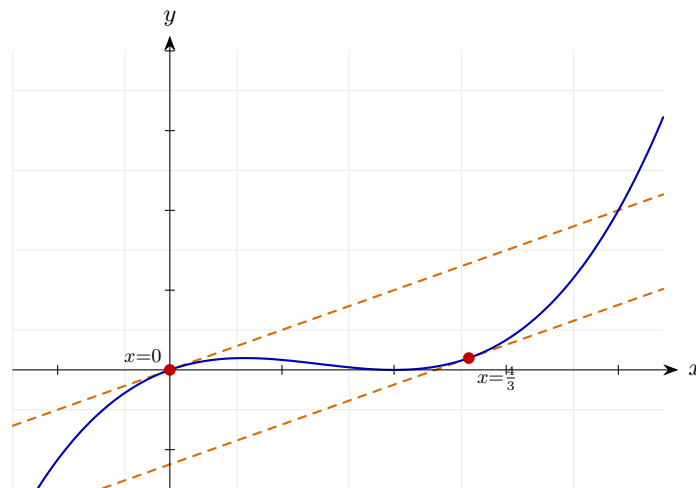
Step 6: Solve.

$$x = 0 \text{ or } x = \frac{4}{3}$$

Two solutions.

$$\text{Final answer: (a) } \frac{dy}{dx} = 3x^2 - 4x + 1 \quad \text{(b) } x = 0 \text{ or } x = \frac{4}{3}.$$

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 66 (d1-066)

Difficulty: Challenge / Marks: 6

Use first principles to show that if $f(x) = ax^2 + bx + c$ then $f'(x) = 2ax + b$, where a, b, c are constants.

Worked solution.

Step 1: Compute $f(x+h)$.

$$a(x+h)^2 + b(x+h) + c = ax^2 + 2axh + ah^2 + bx + bh + c$$

Expand.

Step 2: Subtract $f(x)$.

$$2axh + ah^2 + bh$$

ax^2, bx, c cancel.

Step 3: Divide by h .

$$\frac{2axh + ah^2 + bh}{h} = 2ax + ah + b$$

Simplify.

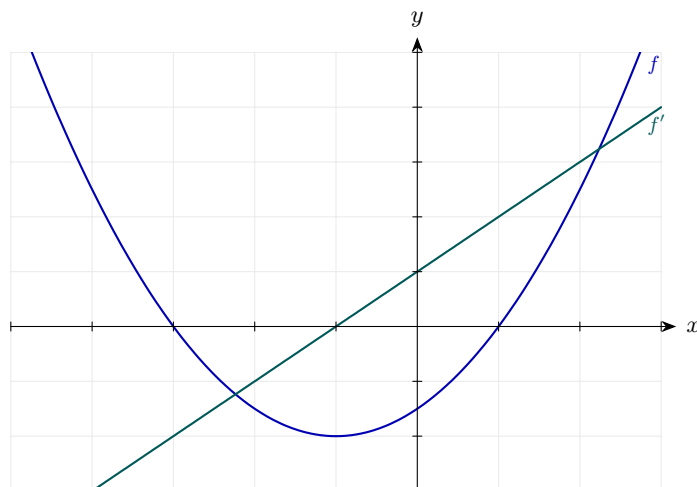
Step 4: Take the limit.

$$f'(x) = \lim_{h \rightarrow 0} (2ax + ah + b) = 2ax + b \checkmark$$

As required.

Final answer: Shown: $f'(x) = 2ax + b$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 67 (d1-067)

Difficulty: Challenge / Marks: 6

The points A and B both lie on the curve $y = x^2$. The chord AB has gradient 5. Given that the x -coordinate of A is a and the x -coordinate of B is $a + h$ with $h \neq 0$, show that $2a + h = 5$. Hence state the gradient of the tangent to the curve at the midpoint of AB .

Worked solution.

Step 1: Gradient of AB .

$$\frac{(a+h)^2 - a^2}{(a+h) - a} = \frac{2ah + h^2}{h} = 2a + h$$

Standard chord-gradient calculation.

Step 2: Set equal to 5.

$$2a + h = 5 \checkmark$$

As required.

Step 3: Midpoint of AB has x -coordinate $a + \frac{h}{2}$.

$$x_M = a + \frac{h}{2}$$

Average of the two x -coordinates.

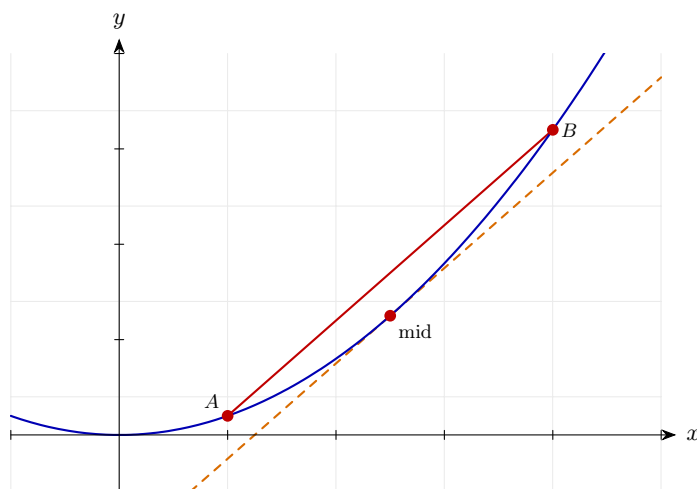
Step 4: Gradient of tangent at midpoint is $2x_M$.

$$\left. \frac{dy}{dx} \right|_{x_M} = 2 \left(a + \frac{h}{2} \right) = 2a + h = 5$$

Using $\frac{dy}{dx} = 2x$ for $y = x^2$. The diagram illustrates that the tangent at the midpoint of AB is parallel to (in fact has the same gradient as) the chord AB .

Final answer: Shown $2a + h = 5$. The tangent at the midpoint has gradient 5 (equal to the chord gradient).

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 68 (d1-068)

Difficulty: Challenge / Marks: 8

The curve C has equation $y = x^2 - 2x + 4$. The point P lies on C and has x -coordinate 3.
 (a) Find the coordinates of P . (b) Use first principles to find the gradient of C at P . (c) Find the equation of the tangent to C at P , in the form $y = mx + c$.

Worked solution.

Step 1: (a) Substitute $x = 3$.

$$y = 9 - 6 + 4 = 7$$

So $P = (3, 7)$.

Step 2: (b) Form $f(3 + h) - f(3)$.

$$(3 + h)^2 - 2(3 + h) + 4 - 7 = 9 + 6h + h^2 - 6 - 2h + 4 - 7 = 4h + h^2$$

Simplify.

Step 3: Difference quotient.

$$\frac{4h + h^2}{h} = 4 + h$$

Divide by h .

Step 4: Take the limit.

$$f'(3) = \lim_{h \rightarrow 0} (4 + h) = 4$$

Gradient at P is 4.

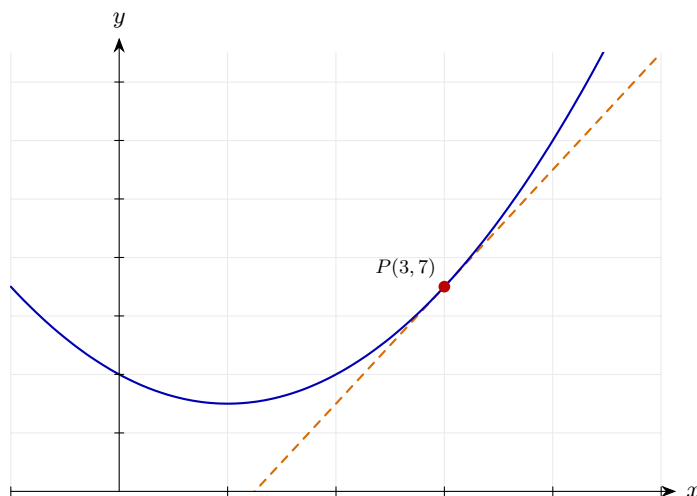
Step 5: (c) Tangent with gradient 4 through $(3, 7)$.

$$y - 7 = 4(x - 3) \Rightarrow y = 4x - 5$$

Point-gradient form, then simplify. The diagram shows curve C and the tangent $y = 4x - 5$ touching at P .

Final answer: (a) $P = (3, 7)$ (b) Gradient = 4 (c) $y = 4x - 5$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 69 (d1-069)

Difficulty: Challenge / Marks: 5

The curve C has equation $y = x^2$. The chord joining the points $P(p, p^2)$ and $Q(q, q^2)$ has gradient m , where $p \neq q$. (a) Show that $m = p + q$. (b) Hence, by letting $q \rightarrow p$, deduce the gradient of the tangent to C at P .

Worked solution.

Step 1: (a) Gradient of chord.

$$m = \frac{q^2 - p^2}{q - p}$$

Gradient formula.

Step 2: Factorise the numerator.

$$\frac{(q - p)(q + p)}{q - p} = q + p \checkmark$$

Difference of two squares; cancel $q - p \neq 0$.

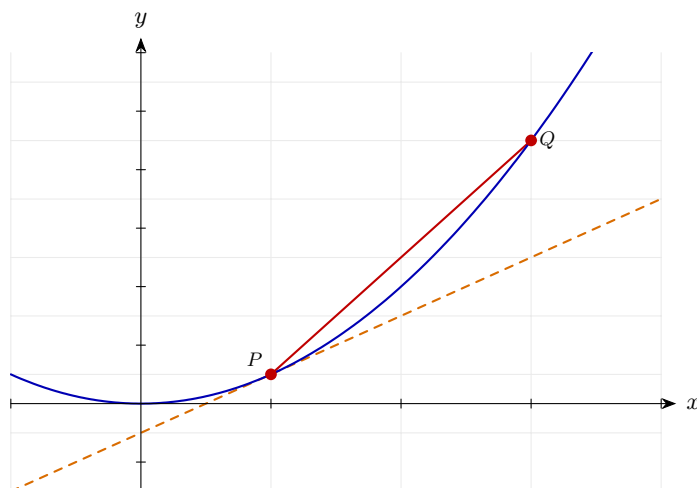
Step 3: (b) Let $q \rightarrow p$.

$$\lim_{q \rightarrow p} (p + q) = 2p$$

Chord becomes tangent. In the diagram, as Q slides along the curve toward P , the chord PQ rotates into the tangent at P , whose gradient is $2p$.

Final answer: (a) Shown $m = p + q$. (b) Tangent gradient at P is $2p$.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.

The Gradient of a Curve 70 (d1-070)

Difficulty: Challenge / Marks: 7

A curve has equation $y = x^3$. The chord joining the points (a, a^3) and $(a + h, (a + h)^3)$, where $h \neq 0$, has gradient m . (a) Show that $m = 3a^2 + 3ah + h^2$. (b) By letting $h \rightarrow 0$, deduce that the gradient of the tangent to the curve at $x = a$ is $3a^2$. (c) Hence find the gradient of the curve at the point where $x = -1$.

Worked solution.

Step 1: (a) Gradient of chord.

$$m = \frac{(a + h)^3 - a^3}{(a + h) - a} = \frac{(a + h)^3 - a^3}{h}$$

Gradient formula.

Step 2: Expand and simplify.

$$(a + h)^3 - a^3 = 3a^2h + 3ah^2 + h^3$$

Binomial expansion.

Step 3: Divide by h .

$$m = 3a^2 + 3ah + h^2 \checkmark$$

As required.

Step 4: (b) Let $h \rightarrow 0$.

$$\lim_{h \rightarrow 0} (3a^2 + 3ah + h^2) = 3a^2$$

All h terms vanish; chord becomes tangent.

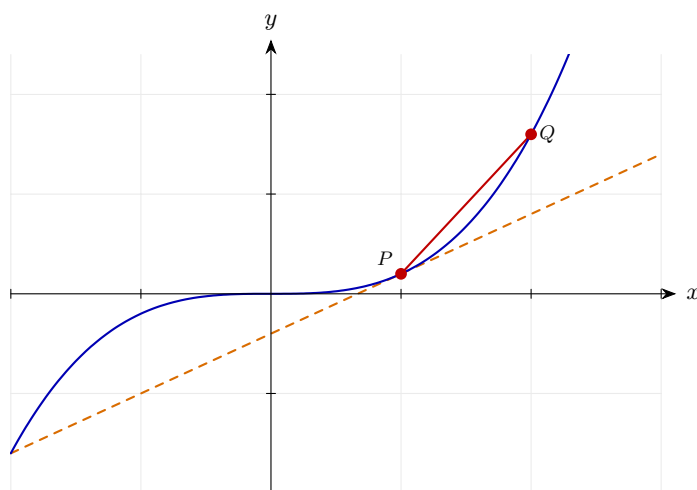
Step 5: (c) Substitute $a = -1$.

$$3(-1)^2 = 3$$

Gradient at $x = -1$ is 3. The diagram shows the curve $y = x^3$ and the tangent at $(-1, -1)$ with gradient 3.

Final answer: (a) Shown. (b) Tangent gradient at $x = a$ is $3a^2$. (c) Gradient at $x = -1$ is 3.

Diagram.



Curve (blue); tangent (dashed orange); chord/derivative as relevant.
